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Prevalence of thrombophilia in women with recurrent early miscarriage, and a systematic review of the literature

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Title: Prevalence of thrombophilia in women with recurrent early miscarriage, and a systematic review of the literature

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Keywords: thrombophilia, antiphospholipid syndrome, recurrent miscarriage, factor V Leiden, prothrombin gene mutation

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Abstract

OBJECTIVE

There are many studies reporting a disproportionally high prevalence of thrombophilia in women with a history of recurrent miscarriage (RM), which has led to over diagnosis and treatment without an improvement in clinical outcomes. The objective of our study was to assess the prevalence of inherited and acquired thrombophilia in a large cohort of women with a history of early RM using internationally agreed diagnostic criteria and inclusion parameters.

SETTING

This is a retrospective cohort study setup in two dedicated tertiary centres for women with recurrent miscarriage in Southwest London and Surrey. We reviewed all the available literature related to causes of recurrent miscarriages. We ascertained the prevalence of thrombophilia in the study population and compared it with historical and published prevalence in the general population.

PARTICIPANTS

1155 women between 2012 and 2017. All patients had three or more first trimester miscarriages and a full thrombophilia screen.

RESULTS

The overall prevalence of thrombophilia in our study population is 9.2% (106/1155) with most cases positive for inherited thrombophilia, 8.1% (94/1155) which is like the general population; Factor V Leiden (FVL) (4.9%; 57/1155) and Prothrombin Gene Mutation (PGM) (2.9%; 34/1155) were the most common thrombophilias. Only 1% (12/1155) tested positive for acquired thrombophilia. Persistent positive lupus anticoagulant (LA) was found in 0.5% (6/1155) and persistent positive anticardiolipin (ACL) antibodies with a value □40 U/ml 0.5% (6/1155) of patients. Both tests were performed a minimum of 12 weeks apart; meeting the revised Sapporo criteria for a diagnosis of antiphospholipid syndrome.

CONCLUSION

Our findings of inherited thrombophilia agree with most of the published literature, therefore do not recommend testing or treatment. When we applied the revised Sapporo criteria, a lower prevalence of acquired thrombophilia was identified. However, the prevalence is higher than the general population, hence we recommend testing for antiphospholipid markers.

STRENGTHS AND LIMITATIONS

- Limitation: the retrospective nature of the study and lack of contemporaneous data from control pregnancies; however, the latter is not a part of routine clinical practice.
- Limitation: Studies included in this review were restricted to women with three or more first trimester miscarriages and those that a minimum of 100 women in the miscarriage arm.
- Strengths 1: the examination of a large cohort of pregnancies with three or more RM in specialist dedicated clinics; accurate and systematic investigation of aetiological causes of early RM including inherited and acquired thrombophilia.
- Strength 2: use of a standardised method of analysis of results; using the same reference range to mitigate errors introduced due to variation in laboratory reporting.
- Strength 3: adherence to the diagnostic criteria of the revised Sapporo criteria.

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STUDY FUNDING/COMPETING INTERESTS
No external funding was used, and no competing interests were declared.

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This study has been approved by NHS Health Research Authority (HRA) (IRAS ID: 281484, REC reference 20/HRA/6376) and is registered with International Prospective Register of systematic Reviews PROSPERO 2020 CRD42020223554.

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Recurrent miscarriage (RM), which affects 1% of couples trying to conceive, is defined as the loss of three or more consecutive pregnancies from the time of conception up to 24 completed weeks of gestation [1]**.** There are differences in recommendations from professional bodies with regards to the definition of RM, with some requiring two or more clinical pregnancies with ultrasound or histological confirmation of pregnancy loss, whereas others requiring three or more losses after a positive pregnancy test with no specification of the need for clinical confirmation [1, 2]**.** These variations in definitions may potentially affect the prevalence of underlying causes of RM in various studies, as the aetiology of biochemical pregnancy loss may be different compared to loss in the second trimester.

Many factors have been studied as possible causes of RM such as anatomical, endocrine, immunological, genetic and thrombophilia (inherited and acquired) disorders. The inherited thrombophilia includes causes such as the presence of Factor V Leiden (FVL) and prothrombin G20210A mutation (PGM), anti-thrombin and protein C and S deficiencies, whereas acquired thrombophilia includes the presence of lupus anticoagulant (LA), anti-cardiolipin (ACL) and anti-beta 2 glycoprotein (B2GP) antibodies. There are many studies reporting a disproportionally high prevalence of thrombophilia in women with a history of RM, which has led to this condition being frequently over diagnosed [3-6]. These studies included women at various gestations and the number of pregnancy losses as well as those identified as having acquired thrombophilia based on values that do not meet the diagnostic criteria. This has led to unnecessary investigations and empirical treatment without improvement in clinical outcomes [7-9]. In contrast, there is evidence from studies suggesting that the prevalence of inherited thrombophilia in women with RM is similar to the general population and that its presence in women with RM is more likely to be an association rather than causation [4, 8-11].

Objective

The objective of our study was to assess the prevalence of inherited and acquired thrombophilia in a large cohort of women with a history of early RM using internationally agreed diagnostic criteria and inclusion parameters. We also systematically reviewed all available literature related to causes of RM.

Setting

This is a retrospective cohort study setup for women with recurrent miscarriage in Epsom and St Helier University Hospitals NHS Trust and the Centre for Reproductive Immunology and Pregnancy, London, and Surrey. We reviewed all the available literature related to causes of recurrent miscarriages. We ascertained the prevalence of thrombophilia in the study population and compared it with historical and published prevalence in the general population.

Participants

1155 women between 2012 and 2017. All couples who have a history of three or more recurrent first trimester losses are eligible for referral to these specialist RM clinics. We record parental demographic information, medical, surgical and gynaecological history and offer comprehensive investigations which include testing for inherited and acquired thrombophilia.

Inclusion and exclusion criteria

The study population included women with a history of three or more first trimester miscarriages and those who had available results for inherited and acquired thrombophilia. We excluded cases that had incomplete results for the thrombophilia screen.

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Ethics

This study has been approved by NHS Health Research Authority (HRA) (IRAS ID: 281484, REC reference 20/HRA/6376).

International Prospective Register of systematic Reviews

This study is registered with International Prospective Register of systematic Reviews PROSPERO 2020 CRD42020223554 where the review process could be accessed.

Data sources and searches

A search of Medline and Embase via the NICE Healthcare Databases Advanced Search interface and a Cochrane database search was performed using Medical Subject Heading (MeSH) terms including exp THROMBOPHILIA/, *"FACTOR V LEIDEN"/, *"ANTIBODIES, ANTICARDIOLIPIN"/ and *"PROTHROMBIN GENE MUTATION"/. The search was very specific and to ensure as much relevant literature on the search topic was retrieved, the major search option was used for the MeSH terms to ensure that the focus of the papers was that particular term. Example terms where the major option was chosen includes *"LUPUS COAGULATION INHIBITOR"/, *"ANTIBODIES and ANTICARDIOLIPIN"/. A limit of English Language, and all papers related to humans and papers dated from 1990 to 2020 were applied. Studies included in this review consisted of observational, case control, prospective and retrospective cohort studies that had women with three or more first trimester miscarriages and those that a minimum of 100 women in the miscarriage arm. Four independent reviewers (AA, MSE, HJ and HS) applied the inclusion criteria to the identified articles from the search and crossed checked the selection process between them. Our search identified 2000 citations; details of article selection are detailed in Figure I. There was a variation in antiphospholipid antibody levels as well as the interval between tests. After applying the diagnostic criteria, the final analysis included 12 papers; ten on inherited and two on acquired thrombophilia.

Outcome measures for inherited thrombophilia and acquired thrombophilia

The investigations included in this paper are homozygous or heterozygous mutations in FVL and PGM. Data of other inherited thrombophilia were collected but not reported due to the very rare incidence in the population.

The investigations for acquired thrombophilia included testing for LA and ACL. We ensured strict criteria for defining a positive antibody test, and therefore by implication, an acquired thrombophilia was present if the result fulfilled the revised Sapporo criteria [12]. A positive LA test was defined as the presence of positive result on two occasions at least 12 weeks apart. The LA screen consisted of a dilute Russell's viper venom time (dRVVT) and an activated partial thromboplastin test (aPTT) test [13]. The normal result for both screening tests was a ratio of ≤ 1.6. If either test was >1.6, a correction was carried out. The test was considered positive if the correction test is >10%. Similarly, a positive ACL antibody test was defined as presence of IgG or IgM antibodies measured using a standardised enzyme linked immunosorbent assay (ELISA) with a level ≥40 U/ml on two occasions at least 12 weeks apart.

Statistical analysis

Data for categorical variables were reported as n (%) and that for continuous variables as median and interquartile ranges (IQR). The summary statistics for prevalence of FVL and PGM were derived for each study in the RM and control groups. These individual study statistics were then combined to obtain a pooled summary estimate which was calculated as a weighted

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average of the individual study estimates in the RM and control groups using both fixed and random-effects model. The fixed-effects model weighs each study by the inverse of its variance and only considers variability in results within studies and not between studies. The random-effects model allows for between-study variability in results by weighting studies using a combination of their own variance and the between-study variance. To determine whether there was a significant difference in the prevalence of thrombophilia between the RM and control groups, we estimated the difference in prevalence (95%CI) between the RM group and control group for each study using the incidence-rate difference (IRD) meta-analysis; the individual summary statistics of difference in prevalence in each study were combined to calculate a weighted pooled estimate for FVL and PGM with IRD meta-analysis. We also estimated the effect size of the significance of difference in prevalence by calculating relative risk (RR) (95%CI) for each study and then a weighted pooled estimate from the RR for each study. Forrest plots of summary statistics for each study and final pooled estimates were constructed using data from the random-effects models. The statistical software package StatsDirect version 2.7.9 (StatsDirect Ltd, Cheshire, United Kingdom) and MedCalc Statistical Software version 16.4.3 (MedCalc Software, Ostend, Belgium) were used for data analysis.

Results

Study population

There was a total of 1,267 women with a history of three or more consecutive early RM in our study population. We excluded cases with missing data and duplicates (n=112) and included data from 1,155 women. The overall prevalence of thrombophilia in our study population was 9.2% (106/1155) of which most cases were positive for inherited thrombophilia 8.1% (94/1155) and 1.0% (12/1155) tested positive for acquired thrombophilia. The median age was 37 years (IQR 33-40) and 66.6% (769/1155) of the cohort had no previous livebirth.

Inherited thrombophilia

The most common inherited thrombophilia was FVL heterozygous mutation, which was noted in 4.8% (55/1155) followed by heterozygous PGM which was identified in 2.9% (34/1155) of patients (Figure IIa and IIb). Only one woman had compound heterozygous for FVL and PGM and one homozygous FVL result. The meta-analysis of proportions demonstrated that the weighted pooled prevalence of FVL in the RM and control group was 5.8% (95%CI: 4.6-7.1%) and 5.2% (95%CI: 3.3-7.5%), respectively (Table I). There was no significant difference in the weighted pooled prevalence of FVL in the RM and control group [0.4 (95%CI:-1.6- 2.3); Z-test=0.396, p=0.692). Similarly, there was no significantly increased risk of RM in women with FVL (RR=1.02 [95%CI: 0.71-1.48]; p=0.906) (Table I, Figure III). The meta-analysis of proportions demonstrated that the weighted pooled prevalence of PGM in the RM and control group was 3.9% (95%CI: 2.7-5.4%) and 2.6% (95%CI: 1.6-3.7%), respectively (Table II). There was no significant difference in the weighted pooled prevalence of PGM in the RM and control group [1.2% (95%CI:-0.4 to 2.8); Z-test=1.463; p=0.143). Similarly, there was no significantly increased risk of RM in women with PGM (RR=1.50 [95%CI: 0.86-2.60]; p=0.154) (Table II, Figure IV).

Acquired thrombophilia results

An initial positive LA test was found in 18 patients, of these, only a third remained positive on repeat testing, giving a LA prevalence of 0.5% (6/1155). A total of 23/1155 (2.0%) patients had two positive ACL tests performed a minimum of 12 weeks apart, based on a lab cut off of ≥10 U/ml. Of these, 2 patients had a value ≥40 U/ml on initial or repeat testing, and 4 had two results with a value >40 U/ml giving an ACL prevalence of 0.5% (6/1155). Details of ACL

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antibody test results are shown in table III. The data concluded the total prevalence of antiphospholipid syndrome is 1.0% (12/1155).

Interpretation and principal findings

The findings of our study demonstrate that the prevalence of inherited thrombophilia in a large cohort of women with a history of early three or more RM is similar to that of the general population, and that the prevalence of acquired thrombophilia is low, albeit higher than the general population. Women and their clinical practitioners should be aware that the contribution of acquired thrombophilia in causing early RM is extremely low, and hence empirical treatment for thrombophilia, such as low dose aspirin and low molecular weight heparin, without establishing an accurate diagnosis should be avoided.

Strengths and limitations

The strengths of our study are the examination of a large cohort of pregnancies with three or more RM in specialist dedicated clinics; accurate and systematic investigation of aetiological causes of early RM including inherited and acquired thrombophilia; use of a standardised method of analysis of results; using the same reference range to mitigate errors introduced due to variation in laboratory reporting; and the adherence to the diagnostic criteria of the revised Sapporo criteria. The limitations of our study are the retrospective nature of the study and lack of contemporaneous data from control pregnancies; however, the latter is not a part of routine clinical practice. Studies included in this review were restricted to women with three or more first trimester miscarriages and those that a minimum of 100 women in the miscarriage arm. The minimum number of 100 women in the study arm could be seen as a limitation.

Comparison with other studies

Our findings are consistent with results of other studies, including large systematic reviews and meta-analyses, which report no significant difference in the prevalence of inherited thrombophilia in women with a history of RM compared to controls [7, 14-16]. In one large case-control study of 1,111 women examining the prevalence of FVL mutation in women with early and late RM compared to controls, the authors reported no significant difference in the prevalence of FVL mutation in early or late RM and controls, which was 3.3%, 3.9% and 4.0%, respectively [14].

Our literature search identified five suitable publications for FVL prevalence and five for PGM prevalence. The prevalence of FVL in the miscarriage control group is 5.2% compared to 7.7% in the general population, while the prevalence of FVL in our miscarriage population is 4.8%. Similarly, the prevalence of PGM in the miscarriage control group is 2.6 % compared to 3.7% in the general population, with our miscarriage population at 2.9%. Therefore, this suggests that there is no significant association of heterozygous FVL or PGM with RM.

In contrast, older studies have reported a high prevalence of inherited thrombophilia in women with RM and therefore advocate routine testing and treatment [17-19]. There is large variation however in the prevalence of FVL and PGM in the general population [20]. This is most likely due to genetic variations with more prevalence seen in the Levant region and Europe compared to the US and the Far East. Despite the widespread practice of screening and treatment, the European Society for Human Reproduction and Embryology [21] and the American Society for Reproductive Medicine guidelines [2] do not recommend screening for inherited thrombophilia. At present the Royal College of Obstetricians and Gynaecologists recommends screening only if women present with pregnancy loss in the second trimester despite little evidence to support this [1].

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Similar to inherited thrombophilia, there is controversial evidence about the association between acquired thrombophilia and RM, with some studies reporting a strong association [22, 23], whereas others question this association [24, 25]. There is also lack of clarity in the literature regarding the presence of anti-phospholipid antibodies, which are found in 1-5% of the general healthy population versus those that are diagnosed with anti-phospholipid syndrome, which is a condition that is present in only 0.05% of the population [26]. A frequently cited study by Rai *et al.* reported a prevalence of antiphospholipid antibodies in women with RM as high as 15% [27]. However, this predated the original Sapporo criteria published in 1999 and hence included levels of antiphospholipid antibodies that were even lower than 10 U/ml, which is the laboratory threshold for diagnosis, as well as less than a 12-week interval between the tests. Applying the revised Sapporo criteria (2006) with the higher laboratory level (40 U/ml) to this study, the number of patients with raised aCL IgG antibodies will reduce from 15 to 4, and the number of raised IgM will reduce from 9 to 3 out of a total of 500 patients giving a prevalence of raised aCL antibodies of 1.4% (7/500) compared to the 5.5% (24/500) quoted in the original paper. With regards to LA the new revised criteria states that levels of DRVVT ratio should be ≥1.2. If this is applied, the number of patients with positive LA will reduce from 46 to 9 which would give a prevalence of 1.8% (9/500) compared to 9.6% (46/500) quoted in the original paper. Hence the revised total prevalence for the Rai *et al.* paper would boast at best a prevalence of 3.2% which most likely would reduce further if the 12-week interval is applied, and not the widely quoted 15%. A more recent study examined the association of anti-phospholipid syndrome and RM using the revised Sapporo criteria for diagnosis and reported that this is present in only 2% of the population of RM [25], while another study reported 0.2% prevalence of LA [24]. Our results are consistent with these findings and reiterate that the prevalence of acquired thrombophilia in women with RM is extremely low. This was highlighted in the recent recommendations from the 14th International Congress on Anti-phospholipid Antibodies Task Force report, which stated that studies reporting the association between anti-phospholipid antibodies and RM are heterogeneous with considerable variation in the definition of RM and in the cut-offs used for defining what constitutes positive antibody levels. They concluded that the association between anti-phospholipid antibodies and RM remains inconclusive and further research is required to establish recommendations [28].

Conclusion

The prevalence of inherited thrombophilia in women with history of RM in the first trimester of pregnancy is similar to that of the general population. The prevalence of acquired thrombophilia although higher than the general population, is still only 1% in our study and between 0.2% and 2% in other good quality publications that adhered to the appropriate diagnostic criteria. The definition of a positive test for thrombophilia should be based on internationally agreed evidence-based criteria and empirical treatment based on either obstetric history or a single test should not be undertaken. Further research is required to investigate the potential impact of treatment options on improving pregnancy outcomes in women with a confirmed diagnosis of acquired thrombophilia.

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Author's roles

HS and AA performed part of the analysis and drafted the manuscript. HS devised the study idea and facilitated data acquisition. MSE, SH, AE, RW assisted with the data collection. AA, MSE, HJ and HS reviewed all the papers from the searches. RA advised on study design and provided statistical analysis. HS, AA, and RA contributed to interpretation of the data and revising the manuscript. The final version was approved by all authors.

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Conflict of interest

The authors report no conflict of interest.

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Data sharing

No additional data available.

Patient and Public Involvement

No patient involved.

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Table 1. Meta-analysis to estimate summary statistics for prevalence of heterozygous Factor V Leiden in the recurrent miscarriage (RM) and control groups. The effect size of the difference in prevalence is expressed as relative risk with 95% confidence intervals

Author	RM group		Control group		Prevalence difference	Relative risk
	Total	n (%,95% CI)	Total	n (%; 95% CI)	% (95% CI)	RR (95% CI)
Rai *et al.,* 2001	904	59 (6.5; 5.0 to 8.3)	150	12 (8.0; 4.2 to 13.6)	-1.5 (-6.0 to 3.0)	0.82 (0.46 to 1.48)
Altintas *et al.,* 2007 [29]	114	9 (7.9; 3.7 to 14.5)	185	13 (7.0; 3.8 to 11.7)	0.9 (-5.5 to 7.2)	1.12 (0.50 to 2.48)
Mougiou *et al.,* 2008 [30]	212	8 (3.8; 1.6 to 7.3)	181	7 (3.9; 1.6 to 7.8)	-0.1 (-4.0 to 3.8)	0.98 (0.37 to 2.55)
Serrano *et al.,* 2011 [31]	100	5 (5.0; 1.6 to 11.3)	100	5 (5.0; 1.6 to 11.3)	-	1.00 (0.32 to 3.14)
Farahmand *et al.,* 2015 [32]	254	11 (4.3; 2.2 to7.6)	350	10 (2.9; 1.4 to 5.2)	1.5 (-1.5 to 4.5)	1.52 (0.67 to 3.43)
Summary statistics						
Fixed effects model	1584	92 (5.9; 4.8 to 7.1)	966	47 (4.9; 3.6 to 6.3)	0.4 (-1.6 to 2.3)	1.03 (0.72 to 1.49)
Random effects model	1584	92 (5.8; 4.6 to 7.1)	966	47 (5.2; 3.3 to 7.5)	0.4 (-1.6 to 2.3)	1.02 (0.71 to 1.48)
Cochrane's Q (p value)		4.401 (0.354)		8.414 (0.078)	1.135 (0.889)	1.456 (0.835)
I^2 (%; 95%CI)		9.1 (0 to 67.3)		52.5 (0 to 80.6)	0 (0 to 64.1)	0 (0 to 64.1)
Bias (p value)		-0.886 (0.524)		3.969 (0.256)	-1.016 (0.542)	1.003 (0.465)

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Table 2. Meta-analysis to estimate summary statistics for prevalence of Prothrombin Gene Mutation (PGM) in the recurrent miscarriage (RM) and control groups

Author	RM group		Control group		Prevalence difference	Relative risk
	Total	n (%,95% CI)	Total	n (%; 95% CI)	% (95% CI)	RR (95% CI)
Altintas *et al.*, 2007 [29]	114	2 (1.8; 0.2 to 6.2)	185	3 (1.6; 0.3 to 4.7)	0.1 (-2.9 to 3.2)	1.08 (0.22 to 5.33)
Mougiou *et al.*, 2008 [30]	212	8 (3.8; 1.6 to 7.3)	181	7 (3.9; 1.6 to 7.8)	-0.1 (-4.0 to 3.8)	0.98 (0.37 to 2.55)
Serrano *et al.*, 2011 [32]	100	3 (3.0; 0.6 to 8.5)	100	1 (1.0; 0.0 to 5.4)	2.0 (-1.9 to 5.9)	3.00 (0.44 to 20.77)
Lino *et al.*, 2014 [33]	106	3 (2.8; 0.6 to 8.0)	98	1 (1.0; 0.0 to 5.6)	1.8 (-2.0 to 5.7)	2.78 (0.40 to 19.21)
Farahmand *et al.*, 2015 [32]	254	13 (5.1; 2.8 to 8.6)	350	10 (2.9; 1.4 to 5.2)	2.3 (-0.9 to 5.4)	1.79 (0.81 to 3.94)
Summary statistics						
Fixed effects model	786	29 (3.9; 2.7 to 5.4)	914	22 (2.6; 1.6 to 3.7)	1.2 (-0.4 to 2.8)	1.52 (0.88 to 2.62)
Random effects model	786	29 (3.9; 2.7 to 5.4)	914	22 (2.6; 1.6 to 3.7)	1.2 (-0.4 to 2.8)	1.50 (0.86 to 2.60)
Cochrane's Q (p value)		2.548 (0.636)		3.293 (0.510)	1.556 (0.817)	1.689 (0.793)
I^2 (%; 95%CI)		0 (0 to 64.1)		0 (0 to 64.1)	0 (0 to 64.1)	0 (0 to 64.1)
Bias (p value)		-1.718 (0.613)		-0.560 (0.841)	0.683 (0.823)	0.473 (0.592)

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Table 3: Results of IgG and IgM testing for anti-cardiolipin (ACL) antibody

IgG/IgM	1st test	2nd test
IgG and IgM results > 10U/ml		
IgM	13	10
IgM	19	27
IgM	22	21
IgM	19	20
IgM	17	18
IgG	28	23
IgG	13	15
IgG	20	19
IgG	15	23
IgG	13	15
IgG	18	32
IgG	21	15
IgG	27	27
IgG	17	20
IgG	21	12
IgG	16	12
IgG	23	25
IgM one result > 40U/ml		
IgM	47	22
IgM	30	48
IgG and IgM one result > 40U/ml		
IgM	45	59
IgM	40	40
IgG	65	48
IgG	59	52

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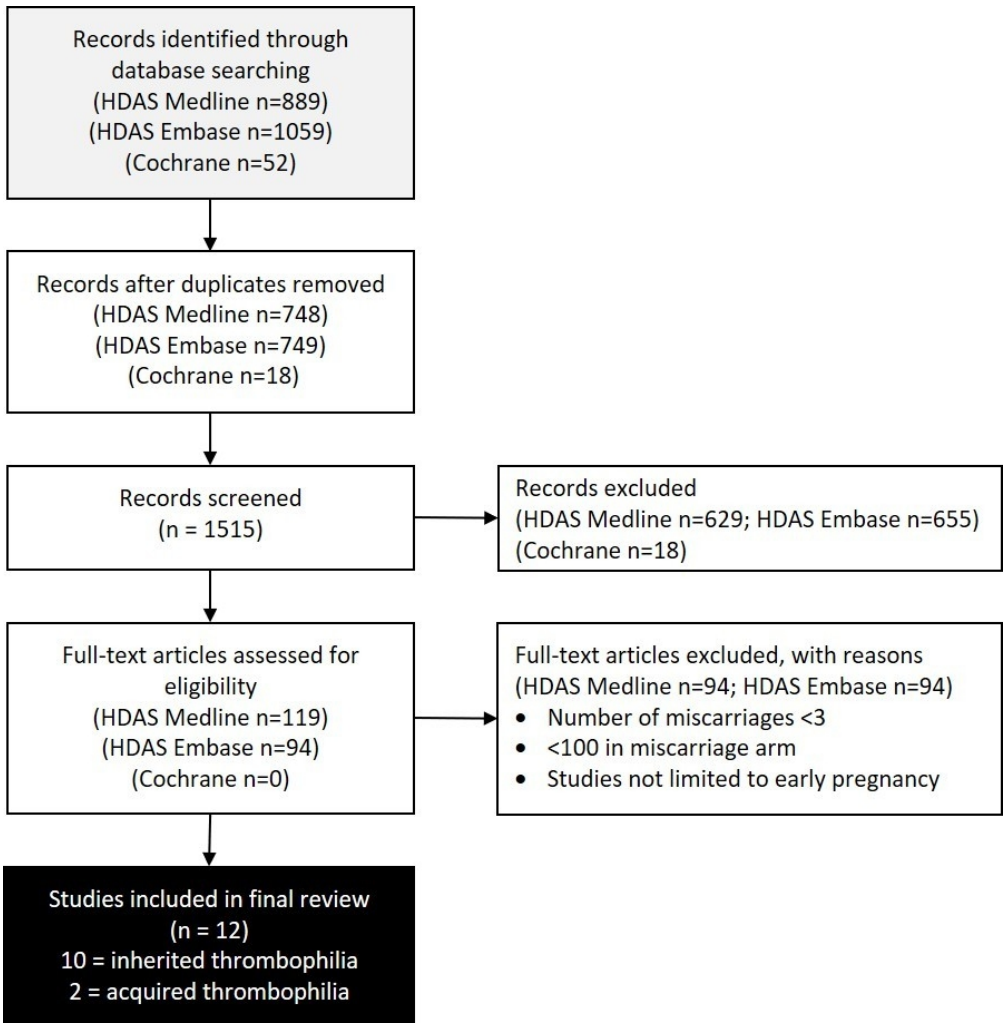
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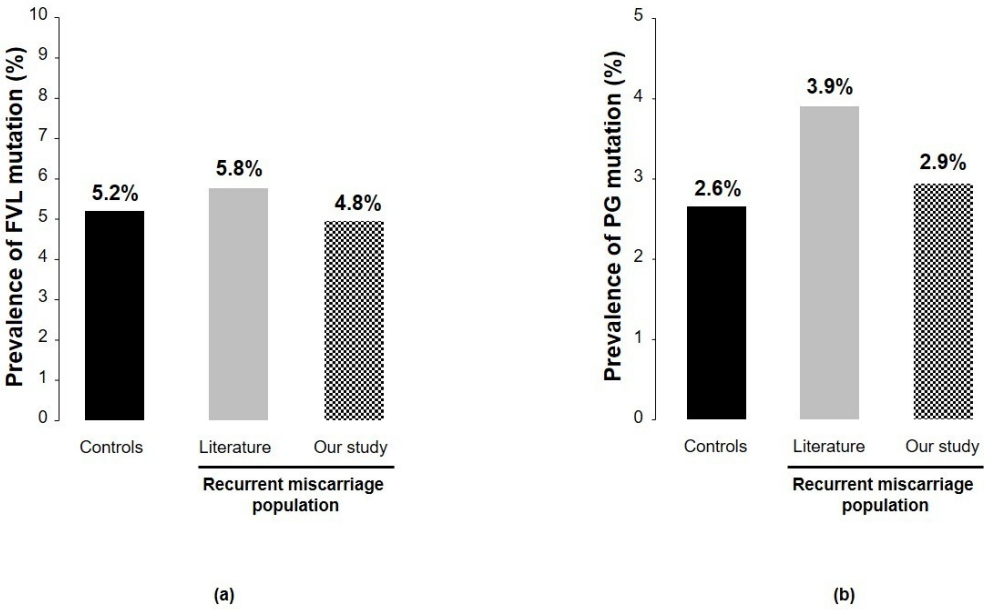
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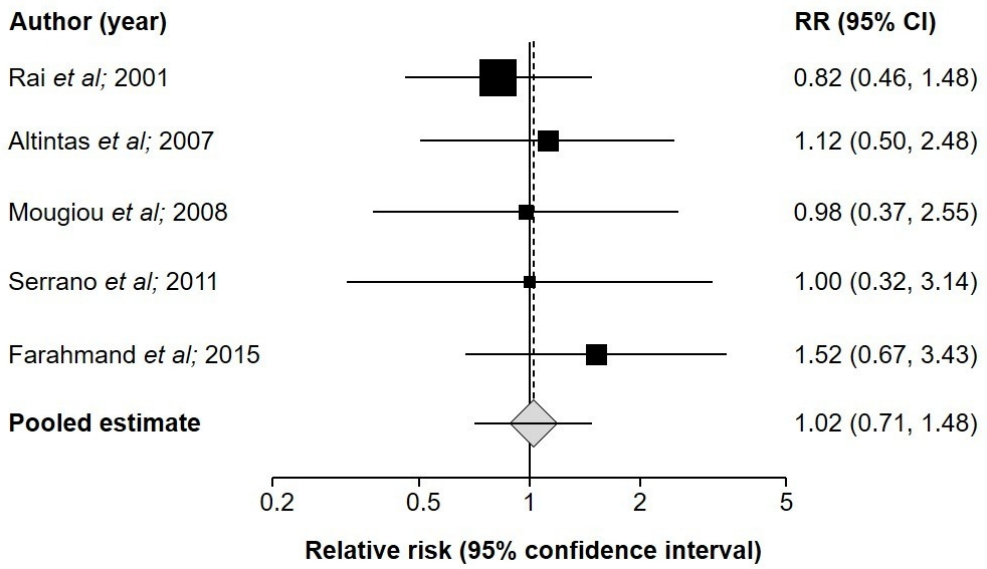
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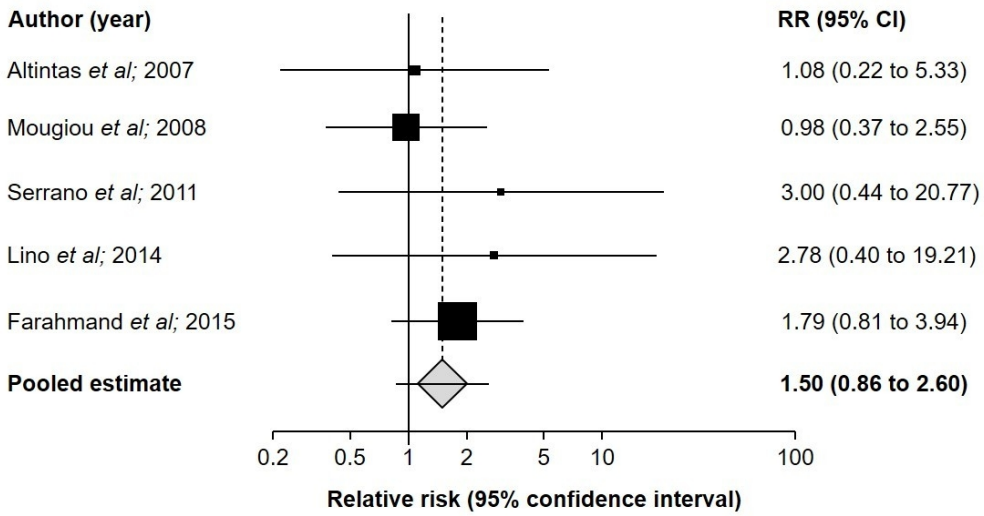
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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Done
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	4
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	5
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	5
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	5
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	5
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	5
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	5
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	5, 6
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	5, 6
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	5, 6, 10, 11
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	5, 6
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	5, 6
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	5, 6
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	5, 6
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	5, 6
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	5, 6, 10, 11
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	5, 6, 10, 11

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	5
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	5
Study characteristics	17	Cite each included study and present its characteristics.	10,11
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	10, 11
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	10, 11
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	5, 6, 10, 11
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	5, 6, 10, 11
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	5, 6, 10, 11
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	5, 6, 10, 11
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	5, 6, 10, 11
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	5, 6, 10, 11
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	7
	23b	Discuss any limitations of the evidence included in the review.	7
	23c	Discuss any limitations of the review processes used.	7
	23d	Discuss implications of the results for practice, policy, and future research.	7, 8
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	5
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	3, 9
Competing interests	26	Declare any competing interests of review authors.	2, 9
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	5

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From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71



PRISMA 2020 Checklist

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STROBE Statement—checklist of items that should be included in reports of observational studies

	Item No	Recommendation	Page No
Title and abstract	1	(*a*) Indicate the study's design with a commonly used term in the title or the abstract	1
		(*b*) Provide in the abstract an informative and balanced summary of what was done and what was found	1
Introduction			
Background/rationale	2	Explain the scientific background and rationale for the investigation being reported	4
Objectives	3	State specific objectives, including any prespecified hypotheses	4
Methods			
Study design	4	Present key elements of study design early in the paper	4
Setting	5	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection	4
Participants	6	(*a*) *Cohort study*—Give the eligibility criteria, and the sources and methods of selection of participants. Describe methods of follow-up *Case-control study*—Give the eligibility criteria, and the sources and methods of case ascertainment and control selection. Give the rationale for the choice of cases and controls *Cross-sectional study*—Give the eligibility criteria, and the sources and methods of selection of participants	4
		(*b*) *Cohort study*—For matched studies, give matching criteria and number of exposed and unexposed *Case-control study*—For matched studies, give matching criteria and the number of controls per case	
Variables	7	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable	5
Data sources/ measurement	8*	For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group	5
Bias	9	Describe any efforts to address potential sources of bias	
Study size	10	Explain how the study size was arrived at	
Quantitative variables	11	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why	
Statistical methods	12	(*a*) Describe all statistical methods, including those used to control for confounding	5, 6
		(*b*) Describe any methods used to examine subgroups and interactions	5, 6
		(*c*) Explain how missing data were addressed	5, 6
		(*d*) *Cohort study*—If applicable, explain how loss to follow-up was addressed *Case-control study*—If applicable, explain how matching of cases and controls was addressed *Cross-sectional study*—If applicable, describe analytical methods taking account of sampling strategy	5, 6
		(*e*) Describe any sensitivity analyses	5, 6

Continued on next page

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Results			
Participants	13*	(a) Report numbers of individuals at each stage of study—eg numbers potentially eligible, examined for eligibility, confirmed eligible, included in the study, completing follow-up, and analysed	6, 7
		(b) Give reasons for non-participation at each stage	
		(c) Consider use of a flow diagram	
Descriptive data	14*	(a) Give characteristics of study participants (eg demographic, clinical, social) and information on exposures and potential confounders	6, 7
		(b) Indicate number of participants with missing data for each variable of interest	6
		(c) *Cohort study*—Summarise follow-up time (eg, average and total amount)	4, 5
Outcome data	15*	*Cohort study*—Report numbers of outcome events or summary measures over time	5
		Case-control study—Report numbers in each exposure category, or summary measures of exposure	
		Cross-sectional study—Report numbers of outcome events or summary measures	
Main results	16	(*a*) Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (eg, 95% confidence interval). Make clear which confounders were adjusted for and why they were included	6, 7
		(*b*) Report category boundaries when continuous variables were categorized	
		(*c*) If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period	
Other analyses	17	Report other analyses done—eg analyses of subgroups and interactions, and sensitivity analyses	
Discussion			
Key results	18	Summarise key results with reference to study objectives	6, 7
Limitations	19	Discuss limitations of the study, taking into account sources of potential bias or imprecision. Discuss both direction and magnitude of any potential bias	7
Interpretation	20	Give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence	7
Generalisability	21	Discuss the generalisability (external validity) of the study results	7, 8
Other information			
Funding	22	Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based	9

*Give information separately for cases and controls in case-control studies and, if applicable, for exposed and unexposed groups in cohort and cross-sectional studies.

Note: An Explanation and Elaboration article discusses each checklist item and gives methodological background and published examples of transparent reporting. The STROBE checklist is best used in conjunction with this article (freely available on the Web sites of PLoS Medicine at http://www.plosmedicine.org/, Annals of Internal Medicine at http://www.annals.org/, and Epidemiology at http://www.epidem.com/). Information on the STROBE Initiative is available at www.strobe-statement.org.

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BMJ Open

Title: Thrombophilia screening in women with recurrent first trimester miscarriage – Is it time to stop testing? – A cohort study and systematic review of the literature

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Secondary Subject Heading:	Haematology (incl blood transfusion), Immunology (including allergy), Reproductive medicine
Keywords:	OBSTETRICS, Anticoagulation < HAEMATOLOGY, Maternal medicine < OBSTETRICS, Reproductive medicine < GYNAECOLOGY

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Title: Thrombophilia screening in women with recurrent first trimester miscarriage – is it time to stop testing? – A cohort study and systematic review of the literature

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Disclosure and conflicts of interest: None

Short title: Thrombophilia investigations in recurrent miscarriages – is it time to stop unnecessary testing and treatment?

Keywords: thrombophilia, antiphospholipid syndrome, recurrent miscarriage, factor V Leiden, prothrombin gene mutation

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ABSTRACT

Objective

There are numerous studies reporting a disproportionally high prevalence of thrombophilia in women with a history of recurrent miscarriage (RM), which has led to over diagnosis and treatment without an improvement in clinical outcomes. The objective of our study was to assess the prevalence of inherited and acquired thrombophilia in a large cohort of women with a history of early RM using internationally agreed diagnostic criteria and inclusion parameters and compare it to the meta-analysis results of existing literature.

Methods

Design

Retrospective cohort study and systematic review of literature

Setting

This is a retrospective cohort study setup in two dedicated tertiary centres for women with recurrent miscarriage in Southwest London and Surrey. We reviewed all the available literature related to causes of recurrent miscarriages. We ascertained the prevalence of thrombophilia in the study population and compared it with historical and published prevalence in the general population.

Participants

1155 women between 2012 and 2017. All patients had three or more first trimester miscarriages and a full thrombophilia screen.

Results

The overall prevalence of thrombophilia in our study population is 9.2% (106/1155) with 8.1% (94/1155) of cases positive for inherited thrombophilia, which is similar to the general population; Factor V Leiden (FVL) (4.9%; 57/1155) and Prothrombin Gene Mutation (PGM) (2.9%; 34/1155) were the most common inherited thrombophilias, while only 1% (12/1155) tested positive for acquired thrombophilia. Persistent positive lupus anticoagulant (LA) was found in 0.5% (6/1155) and persistent positive anticardiolipin (ACL) antibodies with a value ≥40 U/ml was found in 0.5% (6/1155) of patients. Tests for LA/ACL were performed a minimum of 12 weeks apart thus meeting the revised Sapporo criteria for a diagnosis of antiphospholipid syndrome.

Conclusion

The findings of our study demonstrate that the prevalence of inherited thrombophilia is similar in women with recurrent miscarriage to that in the general population. Similarly, the prevalence of acquired thrombophilia, using the revised Sapporo criteria, in the cohort of recurrent miscarriages is similar to that in the general population. Therefore, we do not recommend investigation or treatment of inherited or acquired thrombophilia in women with recurrent miscarriage.

Trial registration number

This study was approved by NHS Health Research Authority (HRA) (IRAS ID: 281484, REC reference:20/HRA/6376) and was registered with International Prospective Register of systematic Reviews (PROSPERO registration number: CRD42020223554).

Strengths and limitations

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- Strength 1: the examination of a large cohort of pregnancies with three or more RM in specialist dedicated clinics; accurate and systematic investigation of aetiological causes of early RM including inherited and acquired thrombophilia.
- Strength 2: use of a standardised method of analysis of results; using the same reference range to mitigate errors introduced due to variation in laboratory reporting such as adherence to the diagnostic criteria of the revised Sapporo criteria.
- Strength 3: systematic review of literature using standard criteria and statistical analysis
- Limitation 1: the retrospective nature of the study and lack of contemporaneous data from control pregnancies; however, the latter is not a part of routine clinical practice.
- Limitation 2: Studies included in this review were restricted to women with three or more first trimester miscarriages and those that a minimum of 100 women in the miscarriage arm.

Funding statement

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INTRODUCTION

Recurrent miscarriage (RM), which affects 1% of couples trying to conceive, is defined as the loss of three or more consecutive pregnancies from the time of conception up to 24 completed weeks of gestation [1]**.** There are differences in recommendations from professional bodies with regards to the definition of RM, with some requiring two or more clinical pregnancies with ultrasound or histological confirmation of pregnancy loss, whereas others requiring three or more losses after a positive pregnancy test with no specification of the need for clinical confirmation [1, 2]**.** These variations in definitions may potentially affect the prevalence of underlying causes of RM in various studies, as the aetiology of biochemical pregnancy loss may be different compared to loss in the second trimester.

Many factors have been studied as possible causes of RM such as anatomical, endocrine, immunological, genetic and thrombophilia (inherited and acquired) disorders. The inherited thrombophilia includes causes such as the presence of Factor V Leiden (FVL) and prothrombin G20210A mutation (PGM), anti-thrombin and protein C and S deficiencies, whereas acquired thrombophilia includes the presence of lupus anticoagulant (LA), anti-cardiolipin (ACL) and anti-beta 2 glycoprotein (B2GP) antibodies. There are many studies reporting a disproportionally high prevalence of thrombophilia in women with a history of RM, which has led to this condition being frequently over diagnosed [3-6]. These studies included women at various gestations and the number of pregnancy losses as well as those identified as having acquired thrombophilia based on values that do not meet the diagnostic criteria. This has led to unnecessary investigations and empirical treatment without improvement in clinical outcomes [7-9]. In contrast, there is evidence from studies suggesting that the prevalence of inherited thrombophilia in women with RM is similar to the general population and that its presence in women with RM is more likely to be an association rather than causation [4, 8-11].

The objective of our study was to assess the prevalence of inherited and acquired thrombophilia in a large cohort of women with a history of early RM using internationally agreed diagnostic criteria and inclusion parameters. We also undertook a systematic review of literature and meta-analysis for studies that reported data regarding inherited and acquired thrombophilia for women with RM and control population.

METHODS

Cohort study

Setting

This is a retrospective cohort study setup for women with recurrent miscarriage in Epsom and St Helier University Hospitals NHS Trust and the Centre for Reproductive Immunology and Pregnancy, London, and Surrey. We reviewed all the available literature related to causes of recurrent miscarriages. We ascertained the prevalence of thrombophilia in the study population and compared it with historical and published prevalence in the general population.

Participants

1155 women between 2012 and 2017. All couples who have a history of three or more recurrent first trimester losses are eligible for referral to these specialist RM clinics. We record parental demographic information, medical, surgical and gynaecological history and offer comprehensive investigations which include testing for inherited and acquired thrombophilia.

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Inclusion and exclusion criteria

The study population included women with a history of three or more first trimester miscarriages and those who had available results for inherited and acquired thrombophilia. We excluded cases that had incomplete results for the thrombophilia screen.

Ethical approval

This study was approved by NHS Health Research Authority (HRA) (IRAS ID: 281484, REC reference:20/HRA/6376).

Patient and public Involvement

No patients were involved in the conduct or design of the study.

Outcome measures for inherited and acquired thrombophilia

The investigations included in this paper are for homozygous or heterozygous mutations in FVL and PGM. Comparison of prevalence of FVL and PGM was made with prevalence in women with RM and controls reported in literature. Data of other inherited thrombophilia were available for a proportion of study cohort study population but were not reported or analysed due to the rare incidence in the population.

The investigations for acquired thrombophilia included testing for LA and ACL. We ensured strict criteria for defining a positive antibody test, and therefore by implication, an acquired thrombophilia was present if the result fulfilled the revised Sapporo criteria [12]. A positive LA test was defined as the presence of positive result on two occasions at least 12 weeks apart. The LA screen consisted of a dilute Russell's viper venom time (dRVVT) and an activated partial thromboplastin test (aPTT) test [13]. The normal result for both screening tests was a ratio of ≤ 1.6. If either test was >1.6, a correction was carried out. The test was considered positive if the correction test is >10%. Similarly, a positive ACL antibody test was defined as presence of IgG or IgM antibodies measured using a standardised enzyme linked immunosorbent assay (ELISA) with a level ≥40 U/ml on two occasions at least 12 weeks apart.

Systematic review and meta-analysis

Study registration

This study was registered with International Prospective Register of systematic Reviews (PROSPERO registration: CRD42020223554).

Data sources and searches

A search of Medline and Embase via the NICE Healthcare Databases Advanced Search interface and a Cochrane database search was performed using Medical Subject Heading (MeSH) terms including exp THROMBOPHILIA/, *"FACTOR V LEIDEN"/, *"ANTIBODIES, ANTICARDIOLIPIN"/ and *"PROTHROMBIN GENE MUTATION"/. The search was very specific and to ensure as much relevant literature on the search topic was retrieved, the major search option was used for the MeSH terms to ensure that the focus of the papers was that particular term. Example terms where the major option was chosen includes *"LUPUS COAGULATION INHIBITOR"/, *"ANTIBODIES and ANTICARDIOLIPIN"/. A limit of English Language, and all papers related to humans and papers dated from 1990 to 2020 were applied. Studies included in this review consisted of observational, case control, prospective and retrospective cohort studies that had women with three or more first trimester miscarriages and those that a minimum of 100 women in the miscarriage arm. Four independent reviewers (AA, MSE, HJ and HS) applied the inclusion criteria to the identified

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articles from the search and crossed checked the selection process between them. Our search identified 2000 citations; details of search criteria are outlined in Supplementary Table 1 and process of article selection are detailed in Figure 1. There was a variation in antiphospholipid antibody levels as well as the interval between tests. After applying the diagnostic criteria, the final analysis included 12 papers; ten on inherited and two on acquired thrombophilia.

Statistical analysis

For all studies in the systematic review, the summary statistics for prevalence of FVL and PGM were derived in the RM and control groups. These individual study statistics were then combined to obtain a pooled summary estimate which was calculated as a weighted average of the individual study estimates in the RM and control groups using both fixed and random-effects model. The fixed-effects model weighs each study by the inverse of its variance and only considers variability in results within studies and not between studies. The random-effects model allows for between-study variability in results by weighting studies using a combination of their own variance and the between-study variance. To determine whether there was a significant difference in the prevalence of thrombophilia between the RM and control groups, we estimated the difference in prevalence (95%CI) between the RM group and control group for each study using the incidence-rate difference (IRD) meta-analysis; the individual summary statistics of difference in prevalence in each study were combined to calculate a weighted pooled estimate for FVL and PGM with IRD meta-analysis. We also estimated the effect size of the significance of difference in prevalence by calculating relative risk (RR) (95%CI) for each study and then a weighted pooled estimate from the RR for each study. Forrest plots of summary statistics for each study and final pooled estimates were constructed using data from the random-effects models.

The heterogeneity between studies was estimated using Cochrane's heterogeneity statistic Q, which was calculated as the weighted sum of squared differences between individual study effects and the pooled effect across studies, with the weights being those used in the pooling method [R]. Inconsistency between study results was assessed using the I^2 statistic, which was calculated as I^2=100% x (Q-df)/Q, where Q is the Cochrane's heterogeneity statistic and df the degrees of freedom. The statistical software package StatsDirect version 2.7.9 (StatsDirect Ltd, Cheshire, United Kingdom) and MedCalc Statistical Software version 16.4.3 (MedCalc Software, Ostend, Belgium) were used for data analysis.

RESULTS

Study population

There was a total of 1,267 women with a history of three or more consecutive early RM in our study population. We excluded cases with missing data and duplicates (n=112) and included data from 1,155 women. The overall prevalence of thrombophilia in our study population was 9.2% (106/1155) of which most cases were positive for inherited thrombophilia 8.1% (94/1155) and 1.0% (12/1155) tested positive for acquired thrombophilia. The median age was 37 years (IQR 33-40) and 66.6% (769/1155) of the cohort had no previous livebirth.

Inherited thrombophilia

The most common inherited thrombophilia was FVL heterozygous mutation, which was noted in 4.8% (55/1155) followed by heterozygous PGM which was identified in 2.9% (34/1155) of patients. There was no significant difference in the prevalence of FVL mutation in our study versus that reported in literature in the RM group ({4.8%} 55/1155 vs. {5.8%} 92/1584, respectively; p=0.263) as well as the prevalence of FVL in RM group versus that of control

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population from literature ({4.8%} 55/1155 vs. {4.9%} 47/966, respectively; p=0.354) (Figure 2). Similarly, there was no significant difference in the prevalence of PGM mutation in our study versus that reported in literature in the RM group ({2.9%} 34/1155 vs. {3.7%} 29/786, respectively; p=0.436) as well as the prevalence of PGM in RM group in our study versus that of control population from literature ({2.9%} 34/1155 vs. {2.4%} 22/914, respectively; p=0.541) (Figure 2). Only one woman had compound heterozygous for FVL and PGM and one homozygous FVL result. The meta-analysis of proportions demonstrated that the weighted pooled prevalence of FVL in the RM and control group was 5.8% (95% CI: 4.6-7.1%) and 5.2% (95%CI: 3.3-7.5%), respectively. There was no significant difference in the weighted pooled prevalence of FVL in the RM and control group [0.4 (95% CI: -1.6- 2.3); Z-test=0.396, p=0.692) (Supplementary Table 2). Similarly, there was no significantly increased risk of RM in women with FVL (RR=1.02 [95%CI: 0.71-1.48]; p=0.906) (Supplementary Table 2, Figure 3). The meta-analysis of proportions demonstrated that the weighted pooled prevalence of PGM in the RM and control group was 3.9% (95%CI: 2.7-5.4%) and 2.6% (95%CI: 1.6-3.7%), respectively (Supplementary Table 3). There was no significant difference in the weighted pooled prevalence of PGM in the RM and control group [1.2% (95% CI: -0.4 to 2.8); Z-test=1.463; p=0.143) (Supplementary Table 3, Figure 4). Similarly, there was no significantly increased risk of RM in women with PGM (RR=1.50 [95%CI: 0.86-2.60]; p=0.154) (Figure 4).

Acquired thrombophilia

An initial positive LA test was found in 18 patients, of these, only a third remained positive on repeat testing, giving a LA prevalence of 0.5% (6/1155). A total of 23/1155 (2.0%) patients had two positive ACL tests performed a minimum of 12 weeks apart, based on a laboratory cut-off value of ≥10 U/ml. Of these, 2 patients had a value ≥40 U/ml on initial or repeat testing, and 4 had two results with a value ≥40 U/ml giving an ACL prevalence of 0.5% (6/1155). The data concluded the total prevalence of antiphospholipid syndrome is 1.0% (12/1155).

DISCUSSION

Principal findings of the study

The findings of our study demonstrate that the prevalence of inherited thrombophilia in a large cohort of women with a history of early three or more RM is similar to that of the general population, and that the prevalence of acquired thrombophilia is low and not significantly different than the prevalence reported in the general population. Women and their clinical practitioners should be aware that the contribution of inherited and acquired thrombophilia in causing early RM is extremely low, and hence empirical testing and treatment for thrombophilia, such as low dose aspirin and low molecular weight heparin should not be advised, unless there is unequivocal clinical and laboratory evidence of established disease.

Strengths and limitations

The strengths of our study are the examination of a large cohort of pregnancies with three or more RM in specialist dedicated clinics; accurate and systematic investigation of aetiological causes of early RM including inherited and acquired thrombophilia; use of a standardised method of analysis of results; using the same reference range to mitigate errors introduced due to variation in laboratory reporting; and the adherence to the diagnostic criteria of the revised Sapporo criteria. The limitations of our study are the retrospective nature of the study and lack of contemporaneous data from control pregnancies; however, the latter is not a part of routine clinical practice. Studies included in this review were restricted to women with three or more

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first trimester miscarriages and those with a minimum of 100 women in the miscarriage arm. The minimum number of 100 women in the study arm could be seen as a limitation.

Comparison with other studies

Our findings are consistent with results of other studies, including large systematic reviews and meta-analyses, which report no significant difference in the prevalence of inherited thrombophilia in women with a history of RM compared to controls [7, 14-16]. In one large case-control study of 1,111 women examining the prevalence of FVL mutation in women with early and late RM compared to controls, the authors reported no significant difference in the prevalence of FVL mutation in early or late RM and controls, which was 3.3%, 3.9% and 4.0%, respectively [14].

Our literature search identified five suitable publications for FVL prevalence (14, 17-20) and five for PGM prevalence (17-21). The prevalence of FVL in the miscarriage control group is 5.2% compared to 7.7% in the general population, while the prevalence of FVL in our miscarriage population is 4.8%. Similarly, the prevalence of PGM in the miscarriage control group is 2.6 % compared to 3.7% in the general population, with our miscarriage population at 2.9%. Therefore, this suggests that there is no significant association of heterozygous FVL or PGM with RM.

In contrast, older studies have reported a high prevalence of inherited thrombophilia in women with RM and therefore advocate routine testing and treatment [22-24]. There is large variation however in the prevalence of FVL and PGM in the general population [25]. This is most likely due to genetic variations with more prevalence seen in the Levant region and Europe compared to the US and the Far East. Despite the widespread practice of screening and treatment, the European Society for Human Reproduction and Embryology [26] and the American Society for Reproductive Medicine guidelines [2] do not recommend screening for inherited thrombophilia. At present the Royal College of Obstetricians and Gynaecologists recommends screening only if women present with pregnancy loss in the second trimester despite little evidence to support this [1].

Similar to inherited thrombophilia, there is controversial evidence about the association between acquired thrombophilia and RM, with some studies reporting a strong association [27, 28], whereas others question this association [29, 30]. The prevalence of anti-phospholipid antibodies in the healthy general population is 1-5% [31]. The prevalence of acquired thrombophilia in women with RM in our study population (1%) is similar or lower when compared to healthy pregnant controls in other studies with a prevalence of ACL IgM at 3%, ACL IgG antibodies at 1% [32] and 1.4% of LA and ACL [33] respectively. A frequently cited study by Rai *et al.* reported a prevalence of antiphospholipid antibodies in women with RM as high as 15% [34]. However, this predated the original Sapporo criteria published in 1999 and hence included levels of antiphospholipid antibodies that were even lower than 10 U/ml, which is the laboratory threshold for diagnosis, as well as less than a 12-week interval between the tests. Applying the revised Sapporo criteria (2006) with the higher laboratory level (40 U/ml) to this study, the number of patients with raised aCL IgG antibodies will reduce from 15 to 4, and the number of raised IgM will reduce from 9 to 3 out of a total of 500 patients giving a prevalence of raised aCL antibodies of 1.4% (7/500) compared to the 5.5% (24/500) quoted in the original paper. With regards to LA, the new revised criteria states that levels of DRVVT ratio should be ≥1.2. If this is applied, the number of patients with positive LA will reduce from 46 to 9 which would give a prevalence of 1.8% (9/500) compared to 9.6% (46/500) quoted in the original paper. Hence the revised total prevalence for the Rai *et al.* paper would boast at

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best a prevalence of 3.2% which most likely would reduce further if the 12-week interval is applied, and not the widely quoted 15%. A more recent study examined the association of anti-phospholipid syndrome and RM using the revised Sapporo criteria for diagnosis and reported that this is present in only 2% of the population of RM [30], while another study reported 0.2% prevalence of LA [29].

Our results are consistent with these findings and reiterate that the prevalence of acquired thrombophilia in women with RM is extremely low and is similar to the general population. This was highlighted in the recent recommendations from the 14th International Congress on Anti-phospholipid Antibodies Task Force report, which stated that studies reporting the association between anti-phospholipid antibodies and RM are heterogeneous with considerable variation in the definition of RM and in the cut-offs used for defining what constitutes positive antibody levels. They concluded that the association between anti-phospholipid antibodies and RM remains inconclusive and further research is required to establish recommendations [35].

CONCLUSION

The prevalence of inherited and acquired thrombophilia in women with history of RM in the first trimester of pregnancy is similar to that of the general population. The prevalence of acquired thrombophilia is only 1% in our study and between 0.2% and 2% in other good quality publications that adhered to the appropriate diagnostic criteria, which implies that the prevalence of not just inherited but also acquired thrombophilia in women with a history of RM is similar to that of the general population.

Therefore, we do not recommend routine investigation or treatment of thrombophilia in women with history of RM, and specifically do not recommend empirical thromboprophylaxis with low molecular weight heparin based on obstetric history in the absence of any clinical evidence of thrombophilia. Further research should be carried out into other aetiologies of RM such as immunological factors and other treatment options.

Author's roles

HS and AA performed part of the analysis and drafted the manuscript. HS devised the study idea and facilitated data acquisition. MSE, SH, AE, RW assisted with the data collection. AA, MSE, HJ and HS reviewed all the papers from the searches. RA advised on study design and provided statistical analysis. HS, AA, and RA contributed to interpretation of the data and revising the manuscript. The final version was approved by all authors.

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Conflict of interest

The authors report no conflict of interest.

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Data sharing

No additional data available.

Patient and Public Involvement

No patients were involved in the conduct or design of the study.

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Figure Legends

Figure 1. Flow chart demonstrating process of selection of studies included in the systematic review

Figure 2. Prevalence of Factor V Leiden (FVL) mutation (2a- left) and Prothrombin gene mutation (PGM) (2b- right) in our study in women with recurrent miscarriages (crossed bar) compared to prevalence reported in literature in recurrent miscarriages (grey solid bar) and general population (black solid bar).

Figure 3. Forest plot demonstrating summary statistics (relative risk [95% confidence intervals]) derived from a random effects model for prevalence of Factor V Leiden (FVL) in women with recurrent miscarriages compared to the control population. Study weights derived from random effect model.

Figure 4. Forest plot demonstrating summary statistics (relative risk [95% confidence intervals]) derived from a random effects model for prevalence of Prothrombin gene mutation (PGM) in women with recurrent miscarriages compared to the control population. Study weights derived from random effect model.

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Supplementary Table 1. Search strategy used in the systematic review

Medline database – Search strategy			
#	Database	Search term	Results
1	Medline	exp "ABORTION, HABITUAL"/	8085
2	Medline	("recurrent miscarriage*").ti,ab	2070
3	Medline	(1 OR 2)	8848
4	Medline	exp THROMBOPHILIA/	24961
5	Medline	("acquired thrombophilia").ti,ab	317
6	Medline	*"PROTEIN C DEFICIENCY"/	979
7	Medline	*"PROTEIN S DEFICIENCY"/	750
8	Medline	*"ANTIPHOSPHOLIPID SYNDROME"/	6568
9	Medline	*"LUPUS COAGULATION INHIBITOR"/	1040
10	Medline	*"ANTIBODIES, ANTICARDIOLIPIN"/	1461
11	Medline	*"BETA 2-GLYCOPROTEIN I"/	568
12	Medline	("factor V Leiden").ti,ab	3998
13	Medline	("prothrombin gene mutation").ti,ab	300
14	Medline	(4 OR 5 OR 6 OR 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13)	35905
15	Medline	(3 AND 14)	889
16	Medline	15 [DT 1990-2020] [Languages English]	760
Embase database – Search strategy			
#	**Database**	**Search term**	**Results**
1.	EMBASE	exp "ABORTION, HABITUAL"/	6141
2.	EMBASE	("recurrent miscarriage*").ti,ab	3620
3.	EMBASE	(21 OR 22)	8255
4.	EMBASE	exp THROMBOPHILIA/	13382
5.	EMBASE	("acquired thrombophilia").ti,ab	623
6.	EMBASE	*"PROTEIN C DEFICIENCY"/	954
7.	EMBASE	*"PROTEIN S DEFICIENCY"/	781
8.	EMBASE	*"ANTIPHOSPHOLIPID SYNDROME"/	9102
9.	EMBASE	*"LUPUS COAGULATION INHIBITOR"/	2122
10.	EMBASE	*"ANTIBODIES, ANTICARDIOLIPIN"/	2548
11.	EMBASE	*"BETA 2-GLYCOPROTEIN I"/	1237
12.	EMBASE	("factor V Leiden").ti,ab	6368
13.	EMBASE	("prothrombin gene mutation").ti,ab	578
14.	EMBASE	(24 OR 25 OR 26 OR 27 OR 28 OR 29 OR 30 OR 31 OR 32 OR 33)	31404
15.	EMBASE	(23 AND 34)	1327

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16.	EMBASE	35 [DT 1990-2020] [English language] [Humans]	1059
Cochrane database – Search strategy			
#	**Database**	**Search term**	**Results**
1.	Cochrane	MeSH descriptor: [Abortion, Habitual] 1 tree(s) exploded	
2.	Cochrane	"recurrent miscarriage*"	
3.	Cochrane	#1 OR #2	
4.	Cochrane	MeSH descriptor: [Thrombophilia] explode all trees	
5.	Cochrane	"acquired thrombophilia"	
6.	Cochrane	MeSH descriptor: [Protein C Deficiency] this term only	
7.	Cochrane	MeSH descriptor: [Protein S Deficiency] this term only	
8.	Cochrane	MeSH descriptor: [Antiphospholipid Syndrome] this term only	
9.	Cochrane	MeSH descriptor: [Lupus Coagulation Inhibitor] explode all trees	
10.	Cochrane	MeSH descriptor: [Antibodies, Anticardiolipin] this term only	
11.	Cochrane	MeSH descriptor: [Antibodies, Anticardiolipin] explode all trees	
12.	Cochrane	"factor V Leiden"	
13.	Cochrane	"prothrombin gene mutation*"	
14.	Cochrane	#4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12	
15.	Cochrane	#3 AND #14	

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Supplementary Table 2. Meta-analysis demonstrating the summary statistics for prevalence of heterozygous Factor V Leiden (FVL) in the recurrent miscarriage and control groups with the prevalence difference (95% confidence intervals (CI) and effect size of the prevalence difference, expressed as relative risk (RR) with 95% CI.

Author	Recurrent miscarriage group		Control group		Prevalence difference	Relative risk
	Total	N (%,95% CI)	Total	N (%; 95% CI)	% (95% CI)	RR (95% CI)
Rai *et al.*, 2001 [14]	904	59 (6.5; 5.0-8.3)	150	12 (8.0; 4.2-13.6)	-1.5 (-6.0-3.0)	0.82 (0.46-1.48)
Altintas *et al.*, 2007 [17]	114	9 (7.9; 3.7-14.5)	185	13 (7.0; 3.8-11.7)	0.9 (-5.5-7.2)	1.12 (0.50-2.48)
Mougiou *et al.*, 2008 [18]	212	8 (3.8; 1.6-7.3)	181	7 (3.9; 1.6-7.8)	-0.1 (-4.0-3.8)	0.98 (0.37-2.55)
Serrano *et al.*, 2011 [19]	100	5 (5.0; 1.6-11.3)	100	5 (5.0; 1.6-11.3)	-	1.00 (0.32-3.14)
Farahmand *et al.*, 2015 [20]	254	11 (4.3; 2.2-7.6)	350	10 (2.9; 1.4-5.2)	1.5 (-1.5-4.5)	1.52 (0.67-3.43)
Summary statistics						
Fixed effects model	1584	92 (5.9; 4.8-7.1)	966	47 (4.9; 3.6-6.3)	0.4 (-1.6-2.3)	1.03 (0.72-1.49)
Random effects model	1584	92 (5.8; 4.6-7.1)	966	47 (5.2; 3.3-7.5)	0.4 (-1.6-2.3)	1.02 (0.71-1.48)
Cochrane's Q (p value)		4.401 (0.354)		8.414 (0.078)	1.135 (0.889)	1.456 (0.835)
I^2 statistic (%; 95%CI)		9.1 (0 to 67.3)		52.5 (0 to 80.6)	0 (0 to 64.1)	0 (0 to 64.1)
Bias (p value)		-0.886 (0.524)		3.969 (0.256)	-1.016 (0.542)	1.003 (0.465)

CI: confidence interval; RR: relative risk

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Supplementary Table 3. Meta-analysis demonstrating the summary statistics for prevalence of heterozygous Prothrombin gene mutation (PGM) in the recurrent miscarriage and control groups with the prevalence difference (95% confidence intervals (CI) and effect size of the prevalence difference, expressed as relative risk (RR) with 95% CI.

Author	Recurrent miscarriage group		Control group		Prevalence difference	Relative risk
	Total	n (%,95% CI)	Total	n (%; 95% CI)	% (95% CI)	RR (95% CI)
Altintas *et al.*, 2007 [17]	114	2 (1.8; 0.2-6.2)	185	3 (1.6; 0.3-4.7)	0.1 (-2.9- 3.2)	1.08 (0.22-5.33)
Mougiou *et al.*, 2008 [18]	212	8 (3.8; 1.6-7.3)	181	7 (3.9; 1.6-7.8)	-0.1 (-4.0-3.8)	0.98 (0.37-2.55)
Serrano *et al.*, 2011 [19]	100	3 (3.0; 0.6-8.5)	100	1 (1.0; 0.0-5.4)	2.0 (-1.9-5.9)	3.00 (0.44-20.77)
Lino *et al.*, 2014 [21]	106	3 (2.8; 0.6-8.0)	98	1 (1.0; 0.0-5.6)	1.8 (-2.0-5.7)	2.78 (0.40-19.21)
Farahmand *et al.*, 2015 [20]	254	13 (5.1; 2.8-8.6)	350	10 (2.9; 1.4-5.2)	2.3 (-0.9-5.4)	1.79 (0.81-3.94)
Summary statistics						
Fixed effects model	786	29 (3.9; 2.7-5.4)	914	22 (2.6; 1.6-3.7)	1.2 (-0.4-2.8)	1.52 (0.88-2.62)
Random effects model	786	29 (3.9; 2.7-5.4)	914	22 (2.6; 1.6-3.7)	1.2 (-0.4-2.8)	1.50 (0.86-2.60)
Cochrane's Q (p value)		2.548 (0.636)		3.293 (0.510)	1.556 (0.817)	1.689 (0.793)
I^2 statistic (%; 95%CI)		0 (0 to 64.1)		0 (0 to 64.1)	0 (0 to 64.1)	0 (0 to 64.1)
Bias (p value)		-1.718 (0.613)		-0.560 (0.841)	0.683 (0.823)	0.473 (0.592)

CI: confidence interval; RR: relative risk

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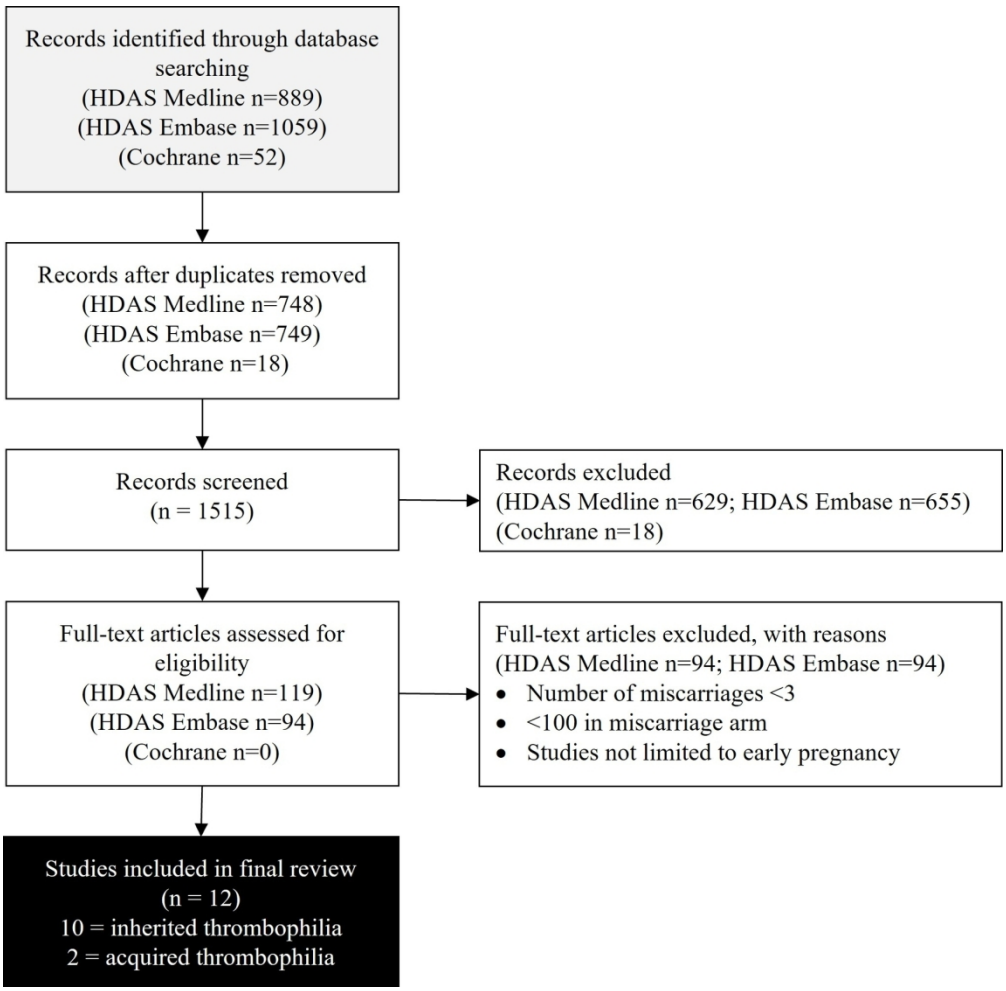


Figure 1. Flow chart demonstrating process of selection of studies included in the systematic review

157x154mm (300 x 300 DPI)

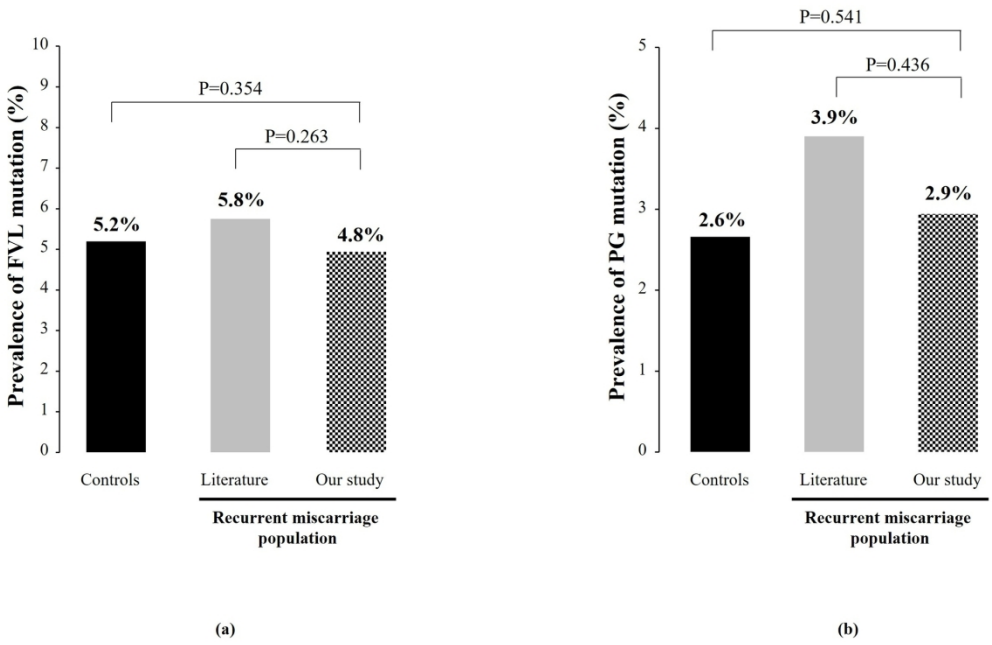


Figure 2. Prevalence of Factor V Leiden (FVL) mutation (2a- left) and Prothrombin gene mutation (PGM) (2b- right) in our study in women with recurrent miscarriages (crossed bar) compared to prevalence reported in literature in recurrent miscarriages (grey solid bar) and general population (black solid bar).

184x120mm (300 x 300 DPI)

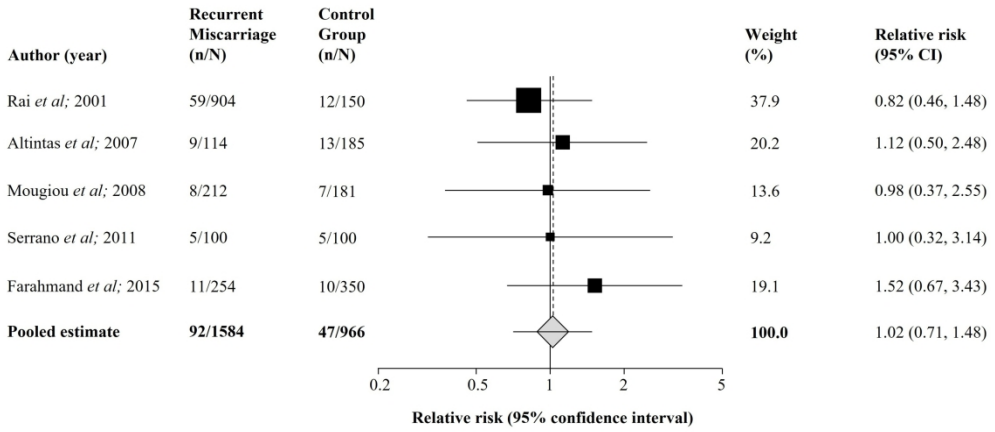


Figure 3. Forest plot demonstrating summary statistics (relative risk [95% confidence intervals]) derived from a random effects model for prevalence of Factor V Leiden (FVL) in women with recurrent miscarriages compared to the control population. Study weights derived from random effect model.

251x110mm (300 x 300 DPI)

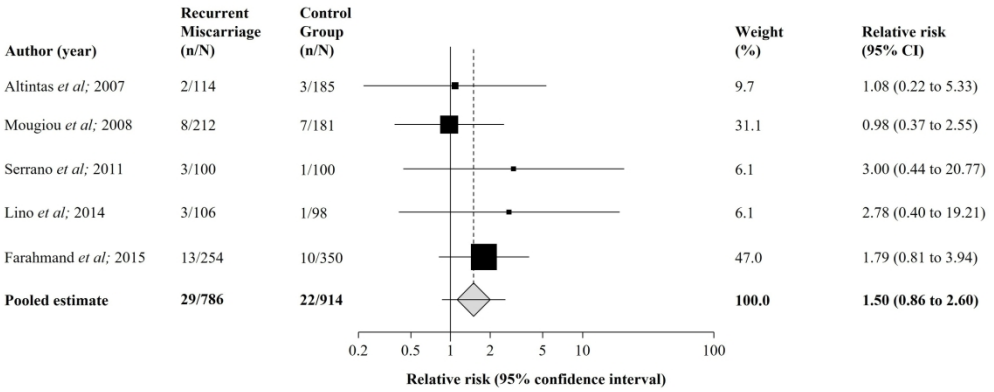


Figure 4. Forest plot demonstrating summary statistics (relative risk [95% confidence intervals]) derived from a random effects model for prevalence of Prothrombin gene mutation (PGM) in women with recurrent miscarriages compared to the control population. Study weights derived from random effect model.

272x108mm (300 x 300 DPI)



PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1, 2, 5
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Done
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	5
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	5, 6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	15
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	5, 6, 9
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	5, 6, 9
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	5, 6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	5, 6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	5, 6, 15
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	5, 6, 7
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	5, 6, 15, 16, 17, 18
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	2, 5, 6, 7, 8
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	17, 18
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	4, 5, 6
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	5, 6
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	6
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	4, 5, 6
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	5, 6, 7

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	6, 7
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	6. 7
Study characteristics	17	Cite each included study and present its characteristics.	11, 12, 13
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	6, 7
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	6, 7, 15, 16, 17, 18
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	6, 7
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	5, 6, 7
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	5, 6, 7
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	5, 6, 7
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	5, 6, 7
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	5, 6, 7
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	7, 8, 9
	23b	Discuss any limitations of the evidence included in the review.	7, 8, 9
	23c	Discuss any limitations of the review processes used.	7, 8, 9
	23d	Discuss implications of the results for practice, policy, and future research.	7, 8, 9
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	2, 5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	5, 6
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	3
Competing interests	26	Declare any competing interests of review authors.	1, 9
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	4, 5, 6

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From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71



PRISMA 2020 Checklist

For more information, visit: http://www.prisma-statement.org/

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STROBE Statement—checklist of items that should be included in reports of observational studies

	Item No	Recommendation	Page No
Title and abstract	1	(*a*) Indicate the study's design with a commonly used term in the title or the abstract	1
		(*b*) Provide in the abstract an informative and balanced summary of what was done and what was found	1
Introduction			
Background/rationale	2	Explain the scientific background and rationale for the investigation being reported	4, 5
Objectives	3	State specific objectives, including any prespecified hypotheses	4, 5
Methods			
Study design	4	Present key elements of study design early in the paper	4, 5
Setting	5	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection	4, 5
Participants	6	(*a*) *Cohort study*—Give the eligibility criteria, and the sources and methods of selection of participants. Describe methods of follow-up *Case-control study*—Give the eligibility criteria, and the sources and methods of case ascertainment and control selection. Give the rationale for the choice of cases and controls *Cross-sectional study*—Give the eligibility criteria, and the sources and methods of selection of participants	4, 5
		(*b*) *Cohort study*—For matched studies, give matching criteria and number of exposed and unexposed *Case-control study*—For matched studies, give matching criteria and the number of controls per case	4, 5
Variables	7	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable	4, 5, 6
Data sources/ measurement	8*	For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group	4, 5, 6
Bias	9	Describe any efforts to address potential sources of bias	17, 18
Study size	10	Explain how the study size was arrived at	N/A
Quantitative variables	11	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why	17, 18
Statistical methods	12	(*a*) Describe all statistical methods, including those used to control for confounding	2, 6, 17, 18
		(*b*) Describe any methods used to examine subgroups and interactions	4, 5, 6
		(*c*) Explain how missing data were addressed	4, 5, 6
		(*d*) *Cohort study*—If applicable, explain how loss to follow-up was addressed *Case-control study*—If applicable, explain how matching of cases and controls was addressed *Cross-sectional study*—If applicable, describe analytical methods taking account of sampling strategy	N/A

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	(e) Describe any sensitivity analyses	4, 5, 6, 17, 18

Continued on next page

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Results			
Participants	13*	(a) Report numbers of individuals at each stage of study—eg numbers potentially eligible, examined for eligibility, confirmed eligible, included in the study, completing follow-up, and analysed	6, 7
		(b) Give reasons for non-participation at each stage	N/A
		(c) Consider use of a flow diagram	N/A
Descriptive data	14*	(a) Give characteristics of study participants (eg demographic, clinical, social) and information on exposures and potential confounders	6, 7
		(b) Indicate number of participants with missing data for each variable of interest	6
		(c) *Cohort study*—Summarise follow-up time (eg, average and total amount)	4, 5
Outcome data	15*	*Cohort study*—Report numbers of outcome events or summary measures over time	4, 5
		Case-control study—Report numbers in each exposure category, or summary measures of exposure	N/A
		Cross-sectional study—Report numbers of outcome events or summary measures	N/A
Main results	16	(*a*) Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (eg, 95% confidence interval). Make clear which confounders were adjusted for and why they were included	6, 7, 8, 9
		(*b*) Report category boundaries when continuous variables were categorized	N/A
		(*c*) If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period	N/A
Other analyses	17	Report other analyses done—eg analyses of subgroups and interactions, and sensitivity analyses	N/A
Discussion			
Key results	18	Summarise key results with reference to study objectives	2, 6, 7
Limitations	19	Discuss limitations of the study, taking into account sources of potential bias or imprecision. Discuss both direction and magnitude of any potential bias	2, 7, 8
Interpretation	20	Give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence	2, 7, 8
Generalisability	21	Discuss the generalisability (external validity) of the study results	2, 4, 8
Other information			
Funding	22	Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based	3

*Give information separately for cases and controls in case-control studies and, if applicable, for exposed and unexposed groups in cohort and cross-sectional studies.

Note: An Explanation and Elaboration article discusses each checklist item and gives methodological background and published examples of transparent reporting. The STROBE checklist is best used in conjunction with this article (freely available on the Web sites of PLoS Medicine at http://www.plosmedicine.org/, Annals of Internal Medicine at http://www.annals.org/, and Epidemiology at http://www.epidem.com/). Information on the STROBE Initiative is available at www.strobe-statement.org.

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Supplementary Table 1. Search strategy used in the systematic review

Medline database – Search strategy			
#	Database	Search term	Results
1	Medline	exp "ABORTION, HABITUAL"/	8085
2	Medline	("recurrent miscarriage*").ti,ab	2070
3	Medline	(1 OR 2)	8848
4	Medline	exp THROMBOPHILIA/	24961
5	Medline	("acquired thrombophilia").ti,ab	317
6	Medline	*"PROTEIN C DEFICIENCY"/	979
7	Medline	*"PROTEIN S DEFICIENCY"/	750
8	Medline	*"ANTIPHOSPHOLIPID SYNDROME"/	6568
9	Medline	*"LUPUS COAGULATION INHIBITOR"/	1040
10	Medline	*"ANTIBODIES, ANTICARDIOLIPIN"/	1461
11	Medline	*"BETA 2-GLYCOPROTEIN I"/	568
12	Medline	("factor V Leiden").ti,ab	3998
13	Medline	("prothrombin gene mutation").ti,ab	300
14	Medline	(4 OR 5 OR 6 OR 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13)	35905
15	Medline	(3 AND 14)	889
16	Medline	15 [DT 1990-2020] [Languages English]	760
Embase database – Search strategy			
#	**Database**	**Search term**	**Results**
1.	EMBASE	exp "ABORTION, HABITUAL"/	6141
2.	EMBASE	("recurrent miscarriage*").ti,ab	3620
3.	EMBASE	(21 OR 22)	8255
4.	EMBASE	exp THROMBOPHILIA/	13382
5.	EMBASE	("acquired thrombophilia").ti,ab	623
6.	EMBASE	*"PROTEIN C DEFICIENCY"/	954
7.	EMBASE	*"PROTEIN S DEFICIENCY"/	781
8.	EMBASE	*"ANTIPHOSPHOLIPID SYNDROME"/	9102
9.	EMBASE	*"LUPUS COAGULATION INHIBITOR"/	2122
10.	EMBASE	*"ANTIBODIES, ANTICARDIOLIPIN"/	2548
11.	EMBASE	*"BETA 2-GLYCOPROTEIN I"/	1237
12.	EMBASE	("factor V Leiden").ti,ab	6368
13.	EMBASE	("prothrombin gene mutation").ti,ab	578
14.	EMBASE	(24 OR 25 OR 26 OR 27 OR 28 OR 29 OR 30 OR 31 OR 32 OR 33)	31404
15.	EMBASE	(23 AND 34)	1327

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16.	EMBASE	35 [DT 1990-2020] [English language] [Humans]	1059
Cochrane database – Search strategy			
#	**Database**	**Search term**	**Results**
1.	Cochrane	MeSH descriptor: [Abortion, Habitual] 1 tree(s) exploded	
2.	Cochrane	"recurrent miscarriage*"	
3.	Cochrane	#1 OR #2	
4.	Cochrane	MeSH descriptor: [Thrombophilia] explode all trees	
5.	Cochrane	"acquired thrombophilia"	
6.	Cochrane	MeSH descriptor: [Protein C Deficiency] this term only	
7.	Cochrane	MeSH descriptor: [Protein S Deficiency] this term only	
8.	Cochrane	MeSH descriptor: [Antiphospholipid Syndrome] this term only	
9.	Cochrane	MeSH descriptor: [Lupus Coagulation Inhibitor] explode all trees	
10.	Cochrane	MeSH descriptor: [Antibodies, Anticardiolipin] this term only	
11.	Cochrane	MeSH descriptor: [Antibodies, Anticardiolipin] explode all trees	
12.	Cochrane	"factor V Leiden"	
13.	Cochrane	"prothrombin gene mutation*"	
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15.	Cochrane	#3 AND #14	

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Bias (p value)		-0.886 (0.524)		3.969 (0.256)	-1.016 (0.542)	1.003 (0.465)

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Author	Recurrent miscarriage group		Control group		Prevalence difference	Relative risk
	Total	n (%,95% CI)	Total	n (%; 95% CI)	% (95% CI)	RR (95% CI)
Altintas *et al.*, 2007 [31]	114	2 (1.8; 0.2-6.2)	185	3 (1.6; 0.3-4.7)	0.1 (-2.9- 3.2)	1.08 (0.22-5.33)
Mougiou *et al.*, 2008 [32]	212	8 (3.8; 1.6-7.3)	181	7 (3.9; 1.6-7.8)	-0.1 (-4.0-3.8)	0.98 (0.37-2.55)
Serrano *et al.*, 2011 [33]	100	3 (3.0; 0.6-8.5)	100	1 (1.0; 0.0-5.4)	2.0 (-1.9-5.9)	3.00 (0.44-20.77)
Lino *et al.*, 2014 [35]	106	3 (2.8; 0.6-8.0)	98	1 (1.0; 0.0-5.6)	1.8 (-2.0-5.7)	2.78 (0.40-19.21)
Farahmand *et al.*, 2015 [34]	254	13 (5.1; 2.8-8.6)	350	10 (2.9; 1.4-5.2)	2.3 (-0.9-5.4)	1.79 (0.81-3.94)
Summary statistics						
Fixed effects model	786	29 (3.9; 2.7-5.4)	914	22 (2.6; 1.6-3.7)	1.2 (-0.4-2.8)	1.52 (0.88-2.62)
Random effects model	786	29 (3.9; 2.7-5.4)	914	22 (2.6; 1.6-3.7)	1.2 (-0.4-2.8)	1.50 (0.86-2.60)
Cochrane's Q (p value)		2.548 (0.636)		3.293 (0.510)	1.556 (0.817)	1.689 (0.793)
I^2 statistic (%; 95%CI)		0 (0 to 64.1)		0 (0 to 64.1)	0 (0 to 64.1)	0 (0 to 64.1)
Bias (p value)		-1.718 (0.613)		-0.560 (0.841)	0.683 (0.823)	0.473 (0.592)

CI: confidence interval; RR: relative risk

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BMJ Open

Title: Thrombophilia screening in women with recurrent first trimester miscarriage – Is it time to stop testing? – A cohort study and systematic review of the literature

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Title: Thrombophilia screening in women with recurrent first trimester miscarriage – is it time to stop testing? – A cohort study and systematic review of the literature

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Short title: Thrombophilia investigations in recurrent miscarriages – is it time to stop unnecessary testing and treatment?

Keywords: thrombophilia, antiphospholipid syndrome, recurrent miscarriage, factor V Leiden, prothrombin gene mutation

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ABSTRACT

Objective

There are numerous studies reporting a disproportionally high prevalence of thrombophilia in women with a history of recurrent miscarriage (RM), which has led to over diagnosis and treatment without an improvement in clinical outcomes. The objective of our study was to assess the prevalence of inherited and acquired thrombophilia in a large cohort of women with a history of early RM using internationally agreed diagnostic criteria and inclusion parameters and compare it to the meta-analysis results of existing literature.

Methods

Design

Retrospective cohort study and systematic review of literature

Setting

This is a retrospective cohort study setup in two dedicated tertiary centres for women with recurrent miscarriage in Southwest London and Surrey. We reviewed all the available literature related to causes of recurrent miscarriages. We ascertained the prevalence of thrombophilia in the study population and compared it with historical and published prevalence in the general population.

Participants

1155 women between 2012 and 2017. All patients had three or more first trimester miscarriages and a full thrombophilia screen.

Results

The overall prevalence of thrombophilia in our study population is 9.2% (106/1155) with 8.1% (94/1155) of cases positive for inherited thrombophilia, which is similar to the general population; Factor V Leiden (FVL) (4.9%; 57/1155) and Prothrombin Gene Mutation (PGM) (2.9%; 34/1155) were the most common inherited thrombophilias, while only 1% (12/1155) tested positive for acquired thrombophilia. Persistent positive lupus anticoagulant (LA) was found in 0.5% (6/1155) and persistent positive anticardiolipin (ACL) antibodies with a value ≥40 U/ml was found in 0.5% (6/1155) of patients. Tests for LA/ACL were performed a minimum of 12 weeks apart thus meeting the revised Sapporo criteria for a diagnosis of antiphospholipid syndrome.

Conclusion

The findings of our study demonstrate that the prevalence of inherited thrombophilia is similar in women with recurrent miscarriage to that in the general population. Similarly, the prevalence of acquired thrombophilia, using the revised Sapporo criteria, in the cohort of recurrent miscarriages is similar to that in the general population. Therefore, we do not recommend investigation or treatment of inherited or acquired thrombophilia in women with recurrent miscarriage.

Trial registration number

This study was approved by NHS Health Research Authority (HRA) (IRAS ID: 281484, REC reference:20/HRA/6376) and was registered with International Prospective Register of systematic Reviews (PROSPERO registration number: CRD42020223554).

Strengths and limitations

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- Strength 1: the examination of a large cohort of pregnancies with three or more RM in specialist dedicated clinics; accurate and systematic investigation of aetiological causes of early RM including inherited and acquired thrombophilia.
- Strength 2: use of a standardised method of analysis of results; using the same reference range to mitigate errors introduced due to variation in laboratory reporting such as adherence to the diagnostic criteria of the revised Sapporo criteria.
- Strength 3: systematic review of literature using standard criteria and statistical analysis
- Limitation 1: the retrospective nature of the study and lack of contemporaneous data from control pregnancies; however, the latter is not a part of routine clinical practice.
- Limitation 2: Studies included in this review were restricted to women with three or more first trimester miscarriages and those that a minimum of 100 women in the miscarriage arm.

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INTRODUCTION

Recurrent miscarriage (RM), which affects 1% of couples trying to conceive, is defined as the loss of three or more consecutive pregnancies from the time of conception up to 24 completed weeks of gestation [1]**.** There are differences in recommendations from professional bodies with regards to the definition of RM, with some requiring two or more clinical pregnancies with ultrasound or histological confirmation of pregnancy loss, whereas others requiring three or more losses after a positive pregnancy test with no specification of the need for clinical confirmation [1, 2]**.** These variations in definitions may potentially affect the prevalence of underlying causes of RM in various studies, as the aetiology of biochemical pregnancy loss may be different compared to loss in the second trimester.

Many factors have been studied as possible causes of RM such as anatomical, endocrine, immunological, genetic and thrombophilia (inherited and acquired) disorders. The inherited thrombophilia includes causes such as the presence of Factor V Leiden (FVL) and prothrombin G20210A mutation (PGM), anti-thrombin and protein C and S deficiencies, whereas acquired thrombophilia includes the presence of lupus anticoagulant (LA), anti-cardiolipin (ACL) and anti-beta 2 glycoprotein (B2GP) antibodies. There are many studies reporting a disproportionally high prevalence of thrombophilia in women with a history of RM, which has led to this condition being frequently over diagnosed [3-6]. These studies included women at various gestations and the number of pregnancy losses as well as those identified as having acquired thrombophilia based on values that do not meet the diagnostic criteria. This has led to unnecessary investigations and empirical treatment without improvement in clinical outcomes [7-9]. In contrast, there is evidence from studies suggesting that the prevalence of inherited thrombophilia in women with RM is similar to the general population and that its presence in women with RM is more likely to be an association rather than causation [4, 8-11].

The objective of our study was to assess the prevalence of inherited and acquired thrombophilia in a large cohort of women with a history of early RM using internationally agreed diagnostic criteria and inclusion parameters. We also undertook a systematic review of literature and meta-analysis for studies that reported data regarding inherited and acquired thrombophilia for women with RM and control population.

METHODS

Cohort study

Setting

This is a retrospective cohort study setup for women with recurrent miscarriage in Epsom and St Helier University Hospitals NHS Trust and the Centre for Reproductive Immunology and Pregnancy, London, and Surrey. We reviewed all the available literature related to causes of recurrent miscarriages. We ascertained the prevalence of thrombophilia in the study population and compared it with historical and published prevalence in the general population.

Participants

1155 women between 2012 and 2017. All couples who have a history of three or more recurrent first trimester losses are eligible for referral to these specialist RM clinics. We record parental demographic information, medical, surgical and gynaecological history and offer comprehensive investigations which include testing for inherited and acquired thrombophilia.

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Inclusion and exclusion criteria

The study population included women with a history of three or more first trimester miscarriages and those who had available results for inherited and acquired thrombophilia. We excluded cases that had incomplete results for the thrombophilia screen.

Ethical approval

This study was approved by NHS Health Research Authority (HRA) (IRAS ID: 281484, REC reference:20/HRA/6376).

Patient and public Involvement

No patients were involved in the conduct or design of the study.

Outcome measures for inherited and acquired thrombophilia

The investigations included in this paper are for homozygous or heterozygous mutations in FVL and PGM. Comparison of prevalence of FVL and PGM was made with prevalence in women with RM and controls reported in literature. Data of other inherited thrombophilia were available for a proportion of study cohort study population but were not reported or analysed due to the rare incidence in the population.

The investigations for acquired thrombophilia included testing for LA and ACL. We ensured strict criteria for defining a positive antibody test, and therefore by implication, an acquired thrombophilia was present if the result fulfilled the revised Sapporo criteria [12]. A positive LA test was defined as the presence of positive result on two occasions at least 12 weeks apart. The LA screen consisted of a dilute Russell's viper venom time (dRVVT) and an activated partial thromboplastin test (aPTT) test [13]. The normal result for both screening tests was a ratio of ≤ 1.6. If either test was >1.6, a correction was carried out. The test was considered positive if the correction test is >10%. Similarly, a positive ACL antibody test was defined as presence of IgG or IgM antibodies measured using a standardised enzyme linked immunosorbent assay (ELISA) with a level ≥40 U/ml on two occasions at least 12 weeks apart.

Systematic review and meta-analysis

Study registration

This study was registered with International Prospective Register of systematic Reviews (PROSPERO registration: CRD42020223554).

Data sources and searches

A search of Medline and Embase via the NICE Healthcare Databases Advanced Search interface and a Cochrane database search was performed using Medical Subject Heading (MeSH) terms including exp THROMBOPHILIA/, *"FACTOR V LEIDEN"/, *"ANTIBODIES, ANTICARDIOLIPIN"/ and *"PROTHROMBIN GENE MUTATION"/. The search was very specific and to ensure as much relevant literature on the search topic was retrieved, the major search option was used for the MeSH terms to ensure that the focus of the papers was that particular term. Example terms where the major option was chosen includes *"LUPUS COAGULATION INHIBITOR"/, *"ANTIBODIES and ANTICARDIOLIPIN"/. A limit of English Language, and all papers related to humans and papers dated from 1990 to 2020 were applied. Studies included in this review consisted of observational, case control, prospective and retrospective cohort studies that had women with three or more first trimester miscarriages and those that a minimum of 100 women in the miscarriage arm. Four independent reviewers (AA, MSE, HJ and HS) applied the inclusion criteria to the identified

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articles from the search and crossed checked the selection process between them. Our search identified 2000 citations; details of search criteria are outlined in Supplementary Table 1 and process of article selection are detailed in Figure 1. There was a variation in antiphospholipid antibody levels as well as the interval between tests. After applying the diagnostic criteria, the final analysis included 12 papers; ten on inherited and two on acquired thrombophilia.

Statistical analysis

For all studies in the systematic review, the summary statistics for prevalence of FVL and PGM were derived in the RM and control groups. These individual study statistics were then combined to obtain a pooled summary estimate which was calculated as a weighted average of the individual study estimates in the RM and control groups using both fixed and random-effects model. The fixed-effects model weighs each study by the inverse of its variance and only considers variability in results within studies and not between studies. The random-effects model allows for between-study variability in results by weighting studies using a combination of their own variance and the between-study variance. To determine whether there was a significant difference in the prevalence of thrombophilia between the RM and control groups, we estimated the difference in prevalence (95%CI) between the RM group and control group for each study using the incidence-rate difference (IRD) meta-analysis; the individual summary statistics of difference in prevalence in each study were combined to calculate a weighted pooled estimate for FVL and PGM with IRD meta-analysis. We also estimated the effect size of the significance of difference in prevalence by calculating relative risk (RR) (95%CI) for each study and then a weighted pooled estimate from the RR for each study. Forrest plots of summary statistics for each study and final pooled estimates were constructed using data from the random-effects models.

The heterogeneity between studies was estimated using Cochrane's heterogeneity statistic Q, which was calculated as the weighted sum of squared differences between individual study effects and the pooled effect across studies, with the weights being those used in the pooling method [R]. Inconsistency between study results was assessed using the I^2 statistic, which was calculated as I^2=100% x (Q-df)/Q, where Q is the Cochrane's heterogeneity statistic and df the degrees of freedom. The statistical software package StatsDirect version 2.7.9 (StatsDirect Ltd, Cheshire, United Kingdom) and MedCalc Statistical Software version 16.4.3 (MedCalc Software, Ostend, Belgium) were used for data analysis.

RESULTS

Study population

There was a total of 1,267 women with a history of three or more consecutive early RM in our study population. We excluded cases with missing data and duplicates (n=112) and included data from 1,155 women. The overall prevalence of thrombophilia in our study population was 9.2% (106/1155) of which most cases were positive for inherited thrombophilia 8.1% (94/1155) and 1.0% (12/1155) tested positive for acquired thrombophilia. The median age was 37 years (IQR 33-40) and 66.6% (769/1155) of the cohort had no previous livebirth.

Inherited thrombophilia

The most common inherited thrombophilia was FVL heterozygous mutation, which was noted in 4.8% (55/1155) followed by heterozygous PGM which was identified in 2.9% (34/1155) of patients. There was no significant difference in the prevalence of FVL mutation in our study versus that reported in literature in the RM group ({4.8%} 55/1155 vs. {5.8%} 92/1584, respectively; p=0.263) as well as the prevalence of FVL in RM group versus that of control

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population from literature ({4.8%} 55/1155 vs. {4.9%} 47/966, respectively; p=0.354) (Figure 2). Similarly, there was no significant difference in the prevalence of PGM mutation in our study versus that reported in literature in the RM group ({2.9%} 34/1155 vs. {3.7%} 29/786, respectively; p=0.436) as well as the prevalence of PGM in RM group in our study versus that of control population from literature ({2.9%} 34/1155 vs. {2.4%} 22/914, respectively; p=0.541) (Figure 2). Only one woman had compound heterozygous for FVL and PGM and one homozygous FVL result. The meta-analysis of proportions demonstrated that the weighted pooled prevalence of FVL in the RM and control group was 5.8% (95% CI: 4.6-7.1%) and 5.2% (95%CI: 3.3-7.5%), respectively. There was no significant difference in the weighted pooled prevalence of FVL in the RM and control group [0.4 (95% CI: -1.6- 2.3); Z-test=0.396, p=0.692) (Supplementary Table 2). Similarly, there was no significantly increased risk of RM in women with FVL (RR=1.02 [95%CI: 0.71-1.48]; p=0.906) (Supplementary Table 2, Figure 3). The meta-analysis of proportions demonstrated that the weighted pooled prevalence of PGM in the RM and control group was 3.9% (95%CI: 2.7-5.4%) and 2.6% (95%CI: 1.6-3.7%), respectively (Supplementary Table 3). There was no significant difference in the weighted pooled prevalence of PGM in the RM and control group [1.2% (95% CI: -0.4 to 2.8); Z-test=1.463; p=0.143) (Supplementary Table 3, Figure 4). Similarly, there was no significantly increased risk of RM in women with PGM (RR=1.50 [95%CI: 0.86-2.60]; p=0.154) (Figure 4).

Acquired thrombophilia

An initial positive LA test was found in 18 patients, of these, only a third remained positive on repeat testing, giving a LA prevalence of 0.5% (6/1155). A total of 23/1155 (2.0%) patients had two positive ACL tests performed a minimum of 12 weeks apart, based on a laboratory cut-off value of ≥10 U/ml. Of these, 2 patients had a value ≥40 U/ml on initial or repeat testing, and 4 had two results with a value ≥40 U/ml giving an ACL prevalence of 0.5% (6/1155). The data concluded the total prevalence of antiphospholipid syndrome is 1.0% (12/1155).

DISCUSSION

Principal findings of the study

The findings of our study demonstrate that the prevalence of inherited thrombophilia in a large cohort of women with a history of early three or more RM is similar to that of the general population, and that the prevalence of acquired thrombophilia is low and not significantly different than the prevalence reported in the general population. Women and their clinical practitioners should be aware that the contribution of inherited and acquired thrombophilia in causing early RM is extremely low, and hence empirical testing and treatment for thrombophilia, such as low dose aspirin and low molecular weight heparin should not be advised, unless there is unequivocal clinical and laboratory evidence of established disease.

Strengths and limitations

The strengths of our study are the examination of a large cohort of pregnancies with three or more RM in specialist dedicated clinics; accurate and systematic investigation of aetiological causes of early RM including inherited and acquired thrombophilia; use of a standardised method of analysis of results; using the same reference range to mitigate errors introduced due to variation in laboratory reporting; and the adherence to the diagnostic criteria of the revised Sapporo criteria. The limitations of our study are the retrospective nature of the study and lack of contemporaneous data from control pregnancies; however, the latter is not a part of routine clinical practice. Studies included in this review were restricted to women with three or more

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first trimester miscarriages and those with a minimum of 100 women in the miscarriage arm. The minimum number of 100 women in the study arm could be seen as a limitation.

Comparison with other studies

Our findings are consistent with results of other studies, including large systematic reviews and meta-analyses, which report no significant difference in the prevalence of inherited thrombophilia in women with a history of RM compared to controls [7, 14-16]. In one large case-control study of 1,111 women examining the prevalence of FVL mutation in women with early and late RM compared to controls, the authors reported no significant difference in the prevalence of FVL mutation in early or late RM and controls, which was 3.3%, 3.9% and 4.0%, respectively [14].

Our literature search identified five suitable publications for FVL prevalence [14, 17-20] and five for PGM prevalence [17-21]. The prevalence of FVL in the miscarriage control group is 5.2% compared to 7.7% in the general population [14, 17-20], while the prevalence of FVL in our miscarriage population is 4.8%. Similarly, the prevalence of PGM in the miscarriage control group is 2.6 % compared to 3.7% in the general population [17-21], with our miscarriage population at 2.9%. Therefore, this suggests that there is no significant association of heterozygous FVL or PGM with RM.

In contrast, older studies have reported a high prevalence of inherited thrombophilia in women with RM and therefore advocate routine testing and treatment [22-24]. There is large variation however in the prevalence of FVL and PGM in the general population [25]. This is most likely due to genetic variations with more prevalence seen in the Levant region and Europe compared to the US and the Far East. Despite the widespread practice of screening and treatment, the European Society for Human Reproduction and Embryology [26] and the American Society for Reproductive Medicine guidelines [2] do not recommend screening for inherited thrombophilia. At present the Royal College of Obstetricians and Gynaecologists recommends screening only if women present with pregnancy loss in the second trimester despite little evidence to support this [1].

Similar to inherited thrombophilia, there is controversial evidence about the association between acquired thrombophilia and RM, with some studies reporting a strong association [27, 28], whereas others question this association [29, 30]. The prevalence of anti-phospholipid antibodies in the healthy general population is 1-5% [31]. The prevalence of acquired thrombophilia in women with RM in our study population (1%) is similar or lower when compared to healthy pregnant controls in other studies with a prevalence of ACL IgM at 3%, ACL IgG antibodies at 1% [32] and 1.4% of LA and ACL [33] respectively. A frequently cited study by Rai *et al.* reported a prevalence of antiphospholipid antibodies in women with RM as high as 15% [34]. However, this predated the original Sapporo criteria published in 1999 and hence included levels of antiphospholipid antibodies that were even lower than 10 U/ml, which is the laboratory threshold for diagnosis, as well as less than a 12-week interval between the tests. Applying the revised Sapporo criteria (2006) with the higher laboratory level (40 U/ml) to this study, the number of patients with raised aCL IgG antibodies will reduce from 15 to 4, and the number of raised IgM will reduce from 9 to 3 out of a total of 500 patients giving a prevalence of raised aCL antibodies of 1.4% (7/500) compared to the 5.5% (24/500) quoted in the original paper. With regards to LA, the new revised criteria states that levels of DRVVT ratio should be ≥1.2. If this is applied, the number of patients with positive LA will reduce from 46 to 9 which would give a prevalence of 1.8% (9/500) compared to 9.6% (46/500) quoted in the original paper. Hence the revised total prevalence for the Rai *et al.* paper would boast at

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best a prevalence of 3.2% which most likely would reduce further if the 12-week interval is applied, and not the widely quoted 15%. A more recent study examined the association of anti-phospholipid syndrome and RM using the revised Sapporo criteria for diagnosis and reported that this is present in only 2% of the population of RM [30], while another study reported 0.2% prevalence of LA [29].

Our results are consistent with these findings and reiterate that the prevalence of acquired thrombophilia in women with RM is extremely low and is similar to the general population. This was highlighted in the recent recommendations from the 14th International Congress on Anti-phospholipid Antibodies Task Force report, which stated that studies reporting the association between anti-phospholipid antibodies and RM are heterogeneous with considerable variation in the definition of RM and in the cut-offs used for defining what constitutes positive antibody levels. They concluded that the association between anti-phospholipid antibodies and RM remains inconclusive and further research is required to establish recommendations [35].

CONCLUSION

The prevalence of inherited and acquired thrombophilia in women with history of RM in the first trimester of pregnancy is similar to that of the general population. The prevalence of acquired thrombophilia is only 1% in our study and between 0.2% and 2% in other good quality publications that adhered to the appropriate diagnostic criteria, which implies that the prevalence of not just inherited but also acquired thrombophilia in women with a history of RM is similar to that of the general population.

Therefore, we do not recommend routine investigation or treatment of thrombophilia in women with history of RM, and specifically do not recommend empirical thromboprophylaxis with low molecular weight heparin based on obstetric history in the absence of any clinical evidence of thrombophilia. Further research should be carried out into other aetiologies of RM such as immunological factors and other treatment options.

Author's roles

HS and AA performed part of the analysis and drafted the manuscript. HS devised the study idea and facilitated data acquisition. MSE, SH, AE, RW assisted with the data collection. AA, MSE, HJ and HS reviewed all the papers from the searches. RA advised on study design and provided statistical analysis. HS, AA, and RA contributed to interpretation of the data and revising the manuscript. The final version was approved by all authors.

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Conflict of interest

The authors report no conflict of interest.

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Data sharing

No additional data available.

Patient and Public Involvement

No patients were involved in the conduct or design of the study.

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Figure Legends

Figure 1. Flow chart demonstrating process of selection of studies included in the systematic review

Figure 2. Prevalence of Factor V Leiden (FVL) mutation (2a- left) and Prothrombin gene mutation (PGM) (2b- right) in our study in women with recurrent miscarriages (crossed bar) compared to prevalence reported in literature in recurrent miscarriages (grey solid bar) and general population (black solid bar).

Figure 3. Forest plot demonstrating summary statistics (relative risk [95% confidence intervals]) derived from a random effects model for prevalence of Factor V Leiden (FVL) in women with recurrent miscarriages compared to the control population. Study weights derived from random effect model.

Figure 4. Forest plot demonstrating summary statistics (relative risk [95% confidence intervals]) derived from a random effects model for prevalence of Prothrombin gene mutation (PGM) in women with recurrent miscarriages compared to the control population. Study weights derived from random effect model.

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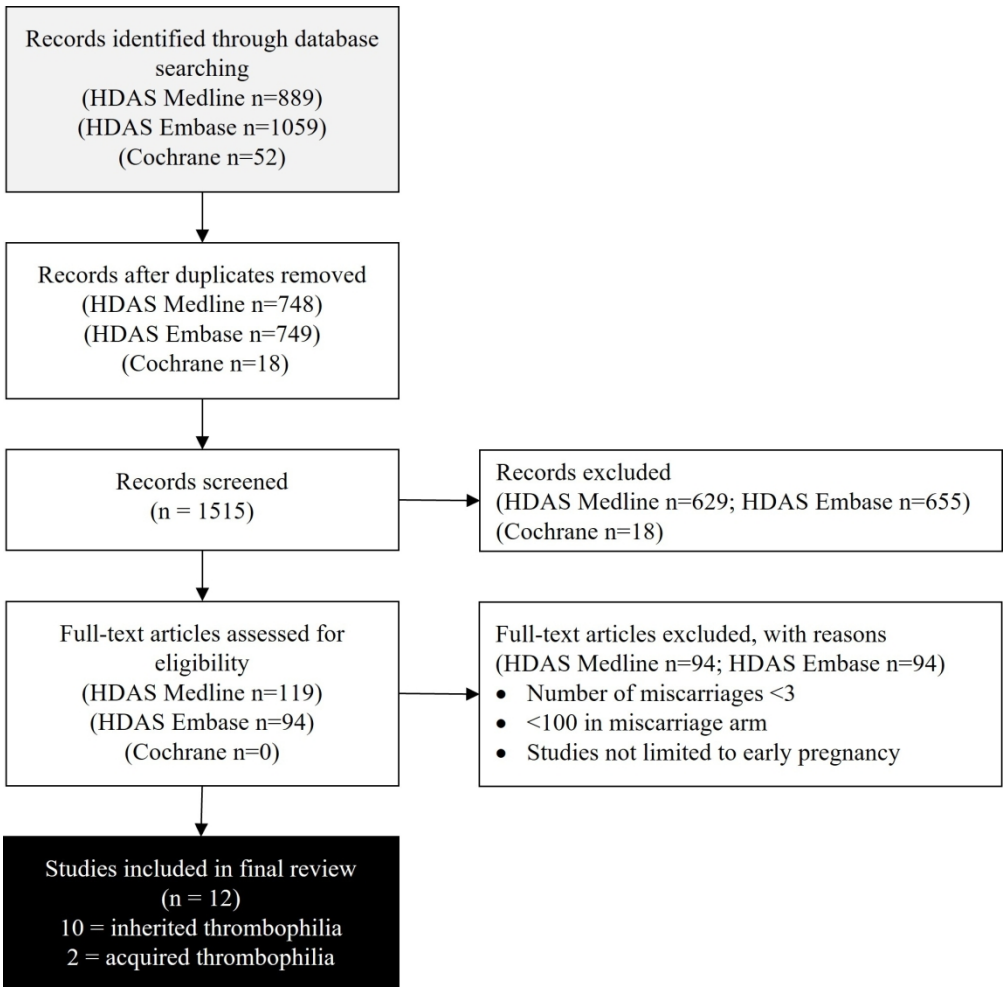


Figure 1. Flow chart demonstrating process of selection of studies included in the systematic review

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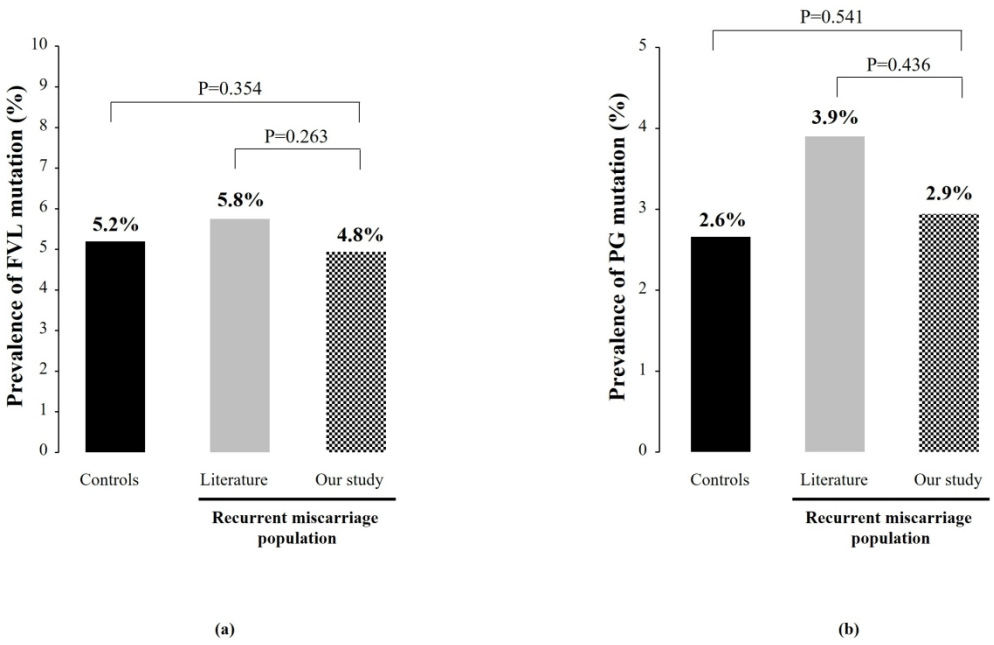


Figure 2. Prevalence of Factor V Leiden (FVL) mutation (2a- left) and Prothrombin gene mutation (PGM) (2b- right) in our study in women with recurrent miscarriages (crossed bar) compared to prevalence reported in literature in recurrent miscarriages (grey solid bar) and general population (black solid bar).

184x120mm (330 x 330 DPI)

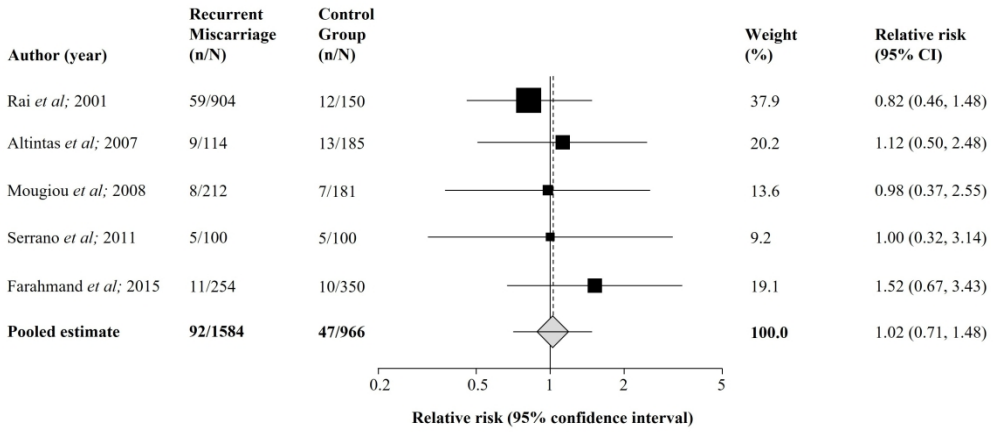


Figure 3. Forest plot demonstrating summary statistics (relative risk [95% confidence intervals]) derived from a random effects model for prevalence of Factor V Leiden (FVL) in women with recurrent miscarriages compared to the control population. Study weights derived from random effect model.

251x110mm (330 x 330 DPI)

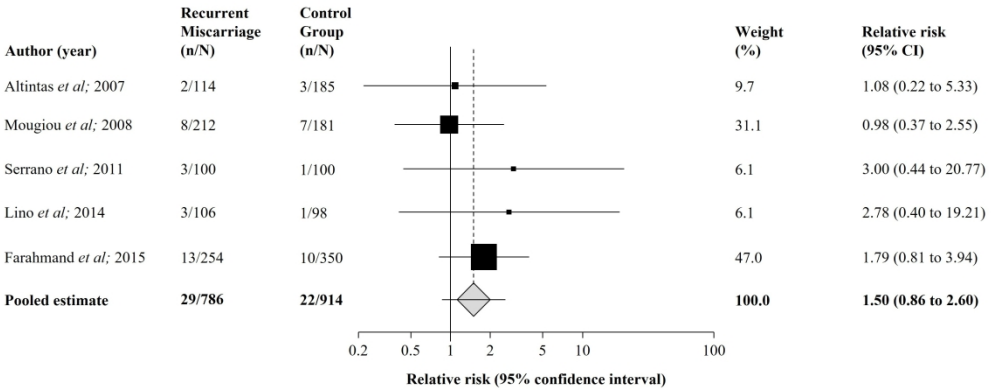


Figure 4. Forest plot demonstrating summary statistics (relative risk [95% confidence intervals]) derived from a random effects model for prevalence of Prothrombin gene mutation (PGM) in women with recurrent miscarriages compared to the control population. Study weights derived from random effect model.

272x108mm (330 x 330 DPI)

Supplementary Table 1. Search strategy used in the systematic review

Medline database – Search strategy			
#	Database	Search term	Results
1	Medline	exp "ABORTION, HABITUAL"/	8085
2	Medline	("recurrent miscarriage*").ti,ab	2070
3	Medline	(1 OR 2)	8848
4	Medline	exp THROMBOPHILIA/	24961
5	Medline	("acquired thrombophilia").ti,ab	317
6	Medline	*"PROTEIN C DEFICIENCY"/	979
7	Medline	*"PROTEIN S DEFICIENCY"/	750
8	Medline	*"ANTIPHOSPHOLIPID SYNDROME"/	6568
9	Medline	*"LUPUS COAGULATION INHIBITOR"/	1040
10	Medline	*"ANTIBODIES, ANTICARDIOLIPIN"/	1461
11	Medline	*"BETA 2-GLYCOPROTEIN I"/	568
12	Medline	("factor V Leiden").ti,ab	3998
13	Medline	("prothrombin gene mutation").ti,ab	300
14	Medline	(4 OR 5 OR 6 OR 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13)	35905
15	Medline	(3 AND 14)	889
16	Medline	15 [DT 1990-2020] [Languages English]	760
Embase database – Search strategy			
#	**Database**	**Search term**	**Results**
1.	EMBASE	exp "ABORTION, HABITUAL"/	6141
2.	EMBASE	("recurrent miscarriage*").ti,ab	3620
3.	EMBASE	(21 OR 22)	8255
4.	EMBASE	exp THROMBOPHILIA/	13382
5.	EMBASE	("acquired thrombophilia").ti,ab	623
6.	EMBASE	*"PROTEIN C DEFICIENCY"/	954
7.	EMBASE	*"PROTEIN S DEFICIENCY"/	781
8.	EMBASE	*"ANTIPHOSPHOLIPID SYNDROME"/	9102
9.	EMBASE	*"LUPUS COAGULATION INHIBITOR"/	2122
10.	EMBASE	*"ANTIBODIES, ANTICARDIOLIPIN"/	2548
11.	EMBASE	*"BETA 2-GLYCOPROTEIN I"/	1237
12.	EMBASE	("factor V Leiden").ti,ab	6368
13.	EMBASE	("prothrombin gene mutation").ti,ab	578
14.	EMBASE	(24 OR 25 OR 26 OR 27 OR 28 OR 29 OR 30 OR 31 OR 32 OR 33)	31404
15.	EMBASE	(23 AND 34)	1327

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16.	EMBASE	35 [DT 1990-2020] [English language] [Humans]	1059

Cochrane database – Search strategy			
#	**Database**	**Search term**	**Results**
1.	Cochrane	MeSH descriptor: [Abortion, Habitual] 1 tree(s) exploded	
2.	Cochrane	"recurrent miscarriage*"	
3.	Cochrane	#1 OR #2	
4.	Cochrane	MeSH descriptor: [Thrombophilia] explode all trees	
5.	Cochrane	"acquired thrombophilia"	
6.	Cochrane	MeSH descriptor: [Protein C Deficiency] this term only	
7.	Cochrane	MeSH descriptor: [Protein S Deficiency] this term only	
8.	Cochrane	MeSH descriptor: [Antiphospholipid Syndrome] this term only	
9.	Cochrane	MeSH descriptor: [Lupus Coagulation Inhibitor] explode all trees	
10.	Cochrane	MeSH descriptor: [Antibodies, Anticardiolipin] this term only	
11.	Cochrane	MeSH descriptor: [Antibodies, Anticardiolipin] explode all trees	
12.	Cochrane	"factor V Leiden"	
13.	Cochrane	"prothrombin gene mutation*"	
14.	Cochrane	#4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12	
15.	Cochrane	#3 AND #14	

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Supplementary Table 2. Meta-analysis demonstrating the summary statistics for prevalence of heterozygous Factor V Leiden (FVL) in the recurrent miscarriage and control groups with the prevalence difference (95% confidence intervals (CI) and effect size of the prevalence difference, expressed as relative risk (RR) with 95% CI.

Author	**Recurrent miscarriage group**		**Control group**		**Prevalence difference**	**Relative risk**
	Total	**N (%,95% CI)**	**Total**	**N (%; 95% CI)**	**% (95% CI)**	**RR (95% CI)**
Rai *et al.*, 2001 [14]	904	59 (6.5; 5.0-8.3)	150	12 (8.0; 4.2-13.6)	-1.5 (-6.0-3.0)	0.82 (0.46-1.48)
Altintas *et al.*, 2007 [31]	114	9 (7.9; 3.7-14.5)	185	13 (7.0; 3.8-11.7)	0.9 (-5.5-7.2)	1.12 (0.50-2.48)
Mougiou *et al.*, 2008 [32]	212	8 (3.8; 1.6-7.3)	181	7 (3.9; 1.6-7.8)	-0.1 (-4.0-3.8)	0.98 (0.37-2.55)
Serrano *et al.*, 2011 [33]	100	5 (5.0; 1.6-11.3)	100	5 (5.0; 1.6-11.3)	-	1.00 (0.32-3.14)
Farahmand *et al.*, 2015 [34]	254	11 (4.3; 2.2-7.6)	350	10 (2.9; 1.4-5.2)	1.5 (-1.5-4.5)	1.52 (0.67-3.43)
Summary statistics						
Fixed effects model	1584	92 (5.9; 4.8-7.1)	966	47 (4.9; 3.6-6.3)	0.4 (-1.6-2.3)	1.03 (0.72-1.49)
Random effects model	1584	92 (5.8; 4.6-7.1)	966	47 (5.2; 3.3-7.5)	0.4 (-1.6-2.3)	1.02 (0.71-1.48)
Cochrane's Q (p value)		4.401 (0.354)		8.414 (0.078)	1.135 (0.889)	1.456 (0.835)
I^2 statistic (%; 95%CI)		9.1 (0 to 67.3)		52.5 (0 to 80.6)	0 (0 to 64.1)	0 (0 to 64.1)
Bias (p value)		-0.886 (0.524)		3.969 (0.256)	-1.016 (0.542)	1.003 (0.465)

CI: confidence interval; RR: relative risk

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Supplementary Table 3. Meta-analysis demonstrating the summary statistics for prevalence of heterozygous Prothrombin gene mutation (PGM) in the recurrent miscarriage and control groups with the prevalence difference (95% confidence intervals (CI) and effect size of the prevalence difference, expressed as relative risk (RR) with 95% CI.

Author	Recurrent miscarriage group		Control group		Prevalence difference	Relative risk
	Total	n (%,95% CI)	Total	n (%; 95% CI)	% (95% CI)	RR (95% CI)
Altintas *et al.*, 2007 [31]	114	2 (1.8; 0.2-6.2)	185	3 (1.6; 0.3-4.7)	0.1 (-2.9- 3.2)	1.08 (0.22-5.33)
Mougiou *et al.*, 2008 [32]	212	8 (3.8; 1.6-7.3)	181	7 (3.9; 1.6-7.8)	-0.1 (-4.0-3.8)	0.98 (0.37-2.55)
Serrano *et al.*, 2011 [33]	100	3 (3.0; 0.6-8.5)	100	1 (1.0; 0.0-5.4)	2.0 (-1.9-5.9)	3.00 (0.44-20.77)
Lino *et al.*, 2014 [35]	106	3 (2.8; 0.6-8.0)	98	1 (1.0; 0.0-5.6)	1.8 (-2.0-5.7)	2.78 (0.40-19.21)
Farahmand *et al.*, 2015 [34]	254	13 (5.1; 2.8-8.6)	350	10 (2.9; 1.4-5.2)	2.3 (-0.9-5.4)	1.79 (0.81-3.94)
Summary statistics						
Fixed effects model	786	29 (3.9; 2.7-5.4)	914	22 (2.6; 1.6-3.7)	1.2 (-0.4-2.8)	1.52 (0.88-2.62)
Random effects model	786	29 (3.9; 2.7-5.4)	914	22 (2.6; 1.6-3.7)	1.2 (-0.4-2.8)	1.50 (0.86-2.60)
Cochrane's Q (p value)		2.548 (0.636)		3.293 (0.510)	1.556 (0.817)	1.689 (0.793)
I^2 statistic (%; 95%CI)		0 (0 to 64.1)		0 (0 to 64.1)	0 (0 to 64.1)	0 (0 to 64.1)
Bias (p value)		-1.718 (0.613)		-0.560 (0.841)	0.683 (0.823)	0.473 (0.592)

CI: confidence interval; RR: relative risk

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	1, 2, 5
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Done
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	5
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	5, 6
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	15
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	5, 6, 9
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	5, 6, 9
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	5, 6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	5, 6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	5, 6, 15
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	5, 6, 7
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	5, 6, 15, 16, 17, 18
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	2, 5, 6, 7, 8
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	17, 18
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	4, 5, 6
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	5, 6
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	6
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	4, 5, 6
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	5, 6, 7

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	6, 7
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	6. 7
Study characteristics	17	Cite each included study and present its characteristics.	11, 12, 13
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	6, 7
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	6, 7, 15, 16, 17, 18
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	6, 7
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	5, 6, 7
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	5, 6, 7
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	5, 6, 7
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	5, 6, 7
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	5, 6, 7
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	7, 8, 9
	23b	Discuss any limitations of the evidence included in the review.	7, 8, 9
	23c	Discuss any limitations of the review processes used.	7, 8, 9
	23d	Discuss implications of the results for practice, policy, and future research.	7, 8, 9
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	2, 5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	5, 6
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	3
Competing interests	26	Declare any competing interests of review authors.	1, 9
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	4, 5, 6

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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STROBE Statement—checklist of items that should be included in reports of observational studies

	Item No	Recommendation	Page No
Title and abstract	1	(*a*) Indicate the study's design with a commonly used term in the title or the abstract	1
		(*b*) Provide in the abstract an informative and balanced summary of what was done and what was found	1
Introduction			
Background/rationale	2	Explain the scientific background and rationale for the investigation being reported	4, 5
Objectives	3	State specific objectives, including any prespecified hypotheses	4, 5
Methods			
Study design	4	Present key elements of study design early in the paper	4, 5
Setting	5	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection	4, 5
Participants	6	(*a*) *Cohort study*—Give the eligibility criteria, and the sources and methods of selection of participants. Describe methods of follow-up *Case-control study*—Give the eligibility criteria, and the sources and methods of case ascertainment and control selection. Give the rationale for the choice of cases and controls *Cross-sectional study*—Give the eligibility criteria, and the sources and methods of selection of participants	4, 5
		(*b*) *Cohort study*—For matched studies, give matching criteria and number of exposed and unexposed *Case-control study*—For matched studies, give matching criteria and the number of controls per case	4, 5
Variables	7	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable	4, 5, 6
Data sources/ measurement	8*	For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group	4, 5, 6
Bias	9	Describe any efforts to address potential sources of bias	17, 18
Study size	10	Explain how the study size was arrived at	N/A
Quantitative variables	11	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why	17, 18
Statistical methods	12	(*a*) Describe all statistical methods, including those used to control for confounding	2, 6, 17, 18
		(*b*) Describe any methods used to examine subgroups and interactions	4, 5, 6
		(*c*) Explain how missing data were addressed	4, 5, 6
		(*d*) *Cohort study*—If applicable, explain how loss to follow-up was addressed *Case-control study*—If applicable, explain how matching of cases and controls was addressed *Cross-sectional study*—If applicable, describe analytical methods taking account of sampling strategy	N/A

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(*e*) Describe any sensitivity analyses	4, 5, 6, 17, 18

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Results			
Participants	13*	(a) Report numbers of individuals at each stage of study—eg numbers potentially eligible, examined for eligibility, confirmed eligible, included in the study, completing follow-up, and analysed	6, 7
		(b) Give reasons for non-participation at each stage	N/A
		(c) Consider use of a flow diagram	N/A
Descriptive data	14*	(a) Give characteristics of study participants (eg demographic, clinical, social) and information on exposures and potential confounders	6, 7
		(b) Indicate number of participants with missing data for each variable of interest	6
		(c) *Cohort study*—Summarise follow-up time (eg, average and total amount)	4, 5
Outcome data	15*	*Cohort study*—Report numbers of outcome events or summary measures over time	4, 5
		Case-control study—Report numbers in each exposure category, or summary measures of exposure	N/A
		Cross-sectional study—Report numbers of outcome events or summary measures	N/A
Main results	16	(*a*) Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (eg, 95% confidence interval). Make clear which confounders were adjusted for and why they were included	6, 7, 8, 9
		(*b*) Report category boundaries when continuous variables were categorized	N/A
		(*c*) If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period	N/A
Other analyses	17	Report other analyses done—eg analyses of subgroups and interactions, and sensitivity analyses	N/A
Discussion			
Key results	18	Summarise key results with reference to study objectives	2, 6, 7
Limitations	19	Discuss limitations of the study, taking into account sources of potential bias or imprecision. Discuss both direction and magnitude of any potential bias	2, 7, 8
Interpretation	20	Give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence	2, 7, 8
Generalisability	21	Discuss the generalisability (external validity) of the study results	2, 4, 8
Other information			
Funding	22	Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based	3

*Give information separately for cases and controls in case-control studies and, if applicable, for exposed and unexposed groups in cohort and cross-sectional studies.

Note: An Explanation and Elaboration article discusses each checklist item and gives methodological background and published examples of transparent reporting. The STROBE checklist is best used in conjunction with this article (freely available on the Web sites of PLoS Medicine at http://www.plosmedicine.org/, Annals of Internal Medicine at http://www.annals.org/, and Epidemiology at http://www.epidem.com/). Information on the STROBE Initiative is available at www.strobe-statement.org.

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